

Hearsay and (non-)endorsement

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Abstract This paper is devoted to what I will call the ‘non-endorsement puzzle’: a distinguished feature of hearsay evidentials such that they allow the speaker to disavow the prejacent. Ever since it was first brought to the forefront of formal semantic research by Martina Faller (2002), the puzzle has been the subject of lively debates in the literature. My goals in this paper are fourfold: (i) to delineate the desiderata for a theory of non-endorsement by placing evidentials in a larger context of speech and attitude reports; (ii) to disentangle the modelling of the core semantics of evidentials from the modelling of the discourse effects of utterances that contain them; (iii) to propose theory-neutral diagnostics of evidential intensionality, thus vindicating a modal analysis for at least some hearsay evidentials; (iv) to argue, building on (Korotkova 2020), that the evidential signal is not conventionally marked as not-at-issue content and that it is at-issue in discourses whose QUD structure allows it (cf. Hunter 2016; Koev 2022 on parenthetical constructions). The existing literature focuses on the modeling question: how to best formulate the *denotation* of hearsay evidentials to make non-endorsement non-contradictory. At the same time, there is an overlooked dimension to the puzzle: the overall *discourse function* of hearsay utterances with overt non-endorsement. As I will argue, the view that evidentials aren’t conventionally not-at-issue allows us to make significant headway in understanding this latter question.

Keywords assertion, cross-linguistic semantics, evidentiality, not-at-issue content, semantics/pragmatics division of labor, speech reports

. Acknowledgments will be added later.

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In Turkish we have a special tense that allows us to distinguish hearsay from what we've seen with our own eyes; when we are relating dreams, fairy tales, or past events we could not have witnessed, we use this tense. It is a useful distinction to make as we "remember" our earliest life experiences, our cradles, our baby carriages, our first steps, all as reported by our parents, stories to which we listen with the same rapt attention we might pay some brilliant tale of some other person. It's a sensation as sweet as seeing ourselves in our dreams, but we pay a heavy price for it. Once imprinted in our minds, other people's reports of what we've done end up mattering more than what we ourselves remember.

—*Istanbul: Memories and the City*, Orhan Pamuk
(translated into English by Maureen Freely)

1 Introduction

This paper is devoted to what I will call the ‘non-endorsement puzzle’: a distinguished feature of hearsay evidentials such that they allow the speaker to disavow the prejacent. Ever since it was first brought to the forefront of formal semantic research by Martina Faller (2002), the puzzle has been the subject of lively debates in the literature. My overarching goal is to place hearsay evidentials in the larger context of speech reports by stressing previously unnoticed parallels between evidentials and communicative predicates both with respect to their conventional meaning and conversational dynamics. Capitalizing on the recent work on not-at-issue meaning and drawing on the familiar diagnostics of intensional environments, I argue that at least some hearsay evidentials are intensional operators and propose that overt non-endorsement is possible in those discourses when both the prejacent and the evidential signal are at-issue.¹

Evidentials are commonly understood as linguistic expressions encoding the source of information for an utterance (Chafe and Nichols 1986; de Haan 2013b,a). Evidence types lexicalized by evidentials include various types of perception, inference and hearsay (Krawczyk 2012; Willett 1988). Starting with Aikhenvald’s (2004) seminal work, typological literature tends to draw a categorical distinction between evidentiality on the one hand and (epistemic) modality on the other. One of the key assumptions is that genuine evidentials—unlike modals—do not encode an agent’s attitude towards the truth of the prejacent (Aikhenvald 2004:4; Aikhenvald 2018:5), at least not as their primary function (see discussion in Boye 2010; Forker 2018). Within formal semantic literature, the relationship between evidentiality and modality is viewed as more nuanced (see Korotkova 2026 for a recent overview). More generally, there is strong support for the idea that the function of evidentials goes beyond indicating one’s information source and is in fact tightly linked to justification and belief. Extensive research on the material of a variety of genetically and geographically unrelated languages has shown that evidentials often signal the speaker’s epistemic stance in discourse (Bergqvist and Grzech 2023; Grzech and Bergqvist 2025) and that many evidentials in fact entail the speaker’s² belief (Murray 2017:17-

1. The literature often frames the puzzle in terms of ‘commitment’. I will stick to the theory-neutral term ‘endorsement’ to avoid wading into the debates on the role of discourse commitments vs. private beliefs in the interpretation of linguistic expressions; see, e.g., Brandom (1983); Krifka (2019) for discussion.

2. The evidence holder is the speaker in unmodified root declaratives, however, like many other perspectival expressions, evidentials can shift in questions and in attitudinal environments. In this paper, I only discuss the behavior of evidentials in matrix declaratives, so for the sake of simplicity I will talk about the speaker’s information state.

21).

The behavior of hearsay evidentials poses a well-known challenge for theories of evidentiality. In many languages, hearsay evidentials do not require the speaker to endorse the pre-jacent, which manifests itself in the availability of explicit disavowals such as ‘I don’t believe ϕ ’ (see overview in [AnderBois 2014](#)). Non-hearsay evidentials, on the other hand, never allow such follow-ups and are thus taken to require some degree of endorsement ([Faller 2002](#); [Martínez Vera 2020](#); [Matthewson et al. 2007](#); [Murray 2010](#), a.o.). Furthermore, even with the so-called indirect evidentials (which are compatible with an inferential and a hearsay interpretation), only hearsay supports the suspension of belief. (1)-(2) illustrate the pattern for Turkish *mİş* ([Şener 2011](#); [Slobin and Akşu 1982](#)) and English *apparently* ([Krawczyk 2012](#)), each of which is compatible with abductive inferences and with hearsay.³ Importantly, the sentences in (2) are felicitous without disavowals in contexts when the speaker didn’t change their mind about the cause of crying. A similar pattern is attested for indirect evidentials in, for example, Bulgarian ([Smirnova 2012](#)) and Georgian ([Korotkova 2012](#)).⁴

(1) **Suspension of belief: Hearsay**

Maria didn’t come to the party yesterday. Ana tells you that she was out of town. But you think that you might have seen her on the street.

a. Maria was **apparently** out of town yesterday, ✓but I don’t believe it.

b. Turkish

Maria dün şehir dışında-y-mış ✓ama ben bun-a inan-mı-yor-um

Maria yesterday town outside-COP-IND but I this-DAT believe-NEG-PRES-1
≈ ‘Maria was out of town yesterday (**I heard**), but I don’t believe it.’

(2) **Suspension of belief: Inference**

You are concerned about Alan’s well-being after he broke up with his long-term partner. After running into him yesterday and seeing that he had red eyes, you decided that he must have been crying but later reconsider. It being the height of pollen season, it must have been just an allergic reaction.

a. Alan **apparently** cried, #but I don’t believe it.

b. Turkish

Alan dün ağla-mış, #ama bu ban-a inandırıcı gel-me-di.

Alan yesterday cry-IND but this I-DAT convincing come-NEG-PST
≈ ‘Alan cried yesterday (**I infer**), #but I don’t find it convincing.’

3. The contrast in the availability of disavowals was first discussed for Turkish by [Şener \(2011\)](#). Earlier literature often reports that *mİş* bans disavowals and that one has to use *güya* ‘allegedly’ in cases like (1b), but my consultants (all below 40) judge such examples felicitous and often volunteer them. I come back to this issue in Section 2.2. Note that like many other languages of the Anatolia-Balkan-Caucasus region, evidential morphology in Turkish also has aspectual interpretations (*mİş* is what [Izvorski \(1997\)](#) calls an ‘evidential perfect’). In what follows, I use temporal frame adverbials to force the evidential interpretation, since such adverbials are incompatible with perfects.

4. Unless indicated otherwise, the data come from my own work with consultants. For languages with non-Latin scripts, I use standard transliteration conventions. Glosses: 1,3 person, ACC accusative, BEN benefactive, COMP complementizer, COND conditional, CONJ conjunction, DAT dative, DEF definite, FUT future, GEN genitive, IND indirect evidential, INDEF indefinite, LOC locative, NEG negation, NMLZ nominalization, NOM nominative, PL plural, PRES present, PROG progressive, PST past, PTCL particle, PTCP participle, REP hearsay (reportative) evidential, SG singular, SUBJ subjunctive, SUP superlative, TOP topic.

The pattern in (1)-(2) has received substantial attention in the literature.⁵ Often taken as an argument against modal approaches to evidentiality (as proposed by [Izvorski 1997](#); [Matthewson et al. 2007](#); [Smirnova 2012](#)), it played a key role in commitment-based theories where evidentials in general are seen as directly interacting with the architecture of discourse and where utterances with hearsay evidentials in particular are analyzed as giving rise to speech acts that differ from regular assertions ([Faller 2002, 2019](#); [Murray 2010, 2014](#)). Much of the existing literature focuses on the modelling question: how to formulate the semantics and discourse updates of hearsay evidentials in a way compatible with disavowals. However, this angle has obscured the variation in the shape of available continuations that signal endorsement and non-endorsement, while those continuations in fact diagnose the variation in the attitudes expressed by those evidentials. The literature also overlooks the discourse function of disavowals, such as why they are uttered and which QUDs they address. As I will argue, we need a theory that offers a principled explanation for the following constellation of facts: (i) non-hearsay evidentials never allow non-endorsement, just like doxastic predicates (e.g., *believe*, *conclude*, *guess*); (ii) hearsay evidentials exhibit lexical variation with respect to endorsement, just like communicative predicates (e.g., *claim* vs. *be right*); (iii) utterances with explicit disavowals, such as (1), have a discourse function distinct from that of hearsay utterances without disavowals, similarly to utterances with communicative predicates. My goals in this paper are fourfold:

- to delineate the desiderata for a theory of (non-)endorsement by placing evidentials in a larger context of speech and attitude reports;
- to disentangle the modelling of the core semantics of evidentials from the modelling of the discourse effects of utterances that contain them;
- to propose theory-neutral diagnostics of evidential intensionality, thus vindicating a modal analysis for at least some hearsay evidentials;
- to argue, building on ([Korotkova 2020](#)), that the evidential signal is not conventionally marked as not-at-issue content and that it is at-issue in discourses with overt non-endorsement (cf. [Hunter 2016](#); [Koev 2022](#) on parenthetical constructions).

I proceed as follows. Section 2 brings in new data that bear on the modeling question. It pays close attention to the linguistic shape of disavowals (something that was previously discussed only in passing) and shows that contrasts found in the evidential domain reflect a more general difference between the language of mental attitudes vs. speech reports. Section 3 shows that at least some hearsay evidentials give rise to intensional effects, namely, they allow contradictions in their scope and block presuppositions of their prejacent. It offers an account of hearsay evidentials as vanilla speech reports, in some cases with additional restrictions regarding the speaker’s attitude towards the prejacent, thus disentangling ‘modality’ from ‘uncertainty’. Section 4 critically evaluates commitment-based approaches to the puzzle and demonstrates that they face several challenges with respect to the modelling question. Section 5 turns to the discourse dynamics of hearsay and argues that evidentials in many ways behave like

5. The data in (1) are sometimes referred to as the evidential version of Moore’s paradox (*#It is raining on Ferenginar, but I don’t believe it*). I maintain that such comparison is confusing: Moore-paradoxical sentences are infelicitous, while hearsay utterances are not, so (1) is not paradoxical to begin with. And even comparing the data in (2) to Moore’s paradox might breed further confusion. Such a comparison suggests an implicit parallel as to the source of infelicity, but that itself warrants further scrutiny. Moore’s infelicities are usually attributed to the pragmatics of assertion ([Gillies 2001](#); [Lawlor and Perry 2008](#); [Stalnaker 2000](#)). However, the jury is still out regarding the badness of disavowals with evidential utterances, including with hearsay (Section 2.2) and in this paper, I will advocate a semantic approach (cf. [Yalcin 2007](#) on epistemic modals).

communicative predicates in discourse, in particular, by having parenthetical and non-parenthetical uses. It refines the question-answer diagnostic of evidential not-at-issueness by taking into account the previously unnoticed PREJACENT UNSETTLEDNESS constraint, shows that some evidentials allow the evidential signal to be the sole main point of an utterance while some other ones mark their prejacent as at-issue, and proposes that overt non-endorsement signals that the evidential signal is at-issue, just like in corresponding discourses with communicative predicates. Importantly, the discourse part of my proposal can be applied to a wide range of evidential expressions, extensional and intensional alike, and is therefore independent of the semantics laid out in Section 3. Section 6 concludes.

An important caveat is in order. Evidentiality, especially in the languages spoken in Europe, doesn't always form a dedicated morphosyntactic category (see detailed discussion in [Bary and Korotkova 2023](#); [Diewald and Smirnova 2010](#)). Evidential distinctions can be encoded via different syntactic means and are often found within modal or tense-aspect grammatical paradigms. For the purposes of this paper, I will understand evidentials very broadly as linguistic expressions that encode semantic notions related to information source, that channel the evidential signal as backgrounded information (at least by default) and that do not express the evidence holder overtly. These properties differentiate evidentials from perception verbs like *hear* and *see* and delineate a loose class of expressions whose semantics and pragmatics will be our main concern throughout this paper.

2 Empirical landscape

2.1 Speech vs. attitude reports

There is a well-known contrast between hearsay vs. non-hearsay evidentials in the possibility to suspend the speaker's belief: based on the languages and constructions that have been studied so far, only hearsay evidentials allow it. [AnderBois \(2014\)](#) makes an important observation that the availability of non-endorsement does not correlate with the morphosyntactic make-up of hearsay evidentials. My goal in this section is to make a stronger claim: the contrast between hearsay vs. non-hearsay evidentials in the possibility of non-endorsement is not arbitrary—it reflects a systematic distinction between speech reports vs. mental attitude reports. For the sake of brevity, I illustrate my claims exclusively with English data, though the general pattern is easily replicated in other languages.

Communicative predicates (when used non-parenthetically, see discussion in Section 5) are often compatible with suspension of belief (3). The same is true for other expressions conveying reported speech (4), including various sentential modifiers like *according to* ([Kratzer 2012](#); [Kaufmann and Kaufmann 2020](#)) or *allegedly*.

(3) COMMUNICATIVE PREDICATES

- a. ✓I **heard/read/was told** that Maria was out of town yesterday, but I don't believe it, I think I saw her.
- b. ✓The papers **claim/report/state/write** that there is a new trade agreement with the Dominion, but I don't believe it/but I know it's just propaganda.

(4) OTHER FORMS OF REPORTED SPEECH

- a. According to Ana, Maria was out of town, but I don't believe it.
- b. [Git] **allegedly** ignores files but actually adds them.⁶

Expressions that encode an agent's doxastic state, on the other hand, always entail that the prejacent is considered at least a live possibility with respect to the index at which it is evaluated. This is easiest to see with doxastic predicates (5). Even weak predicates like *guess* (Dorst and Mandelkern 2023) or hybrid predicates like *doubt* (Anand and Hacquard 2013) require the attitude holder's epistemic state to be compatible with the prejacent. The same is true for other expressions that make reference to beliefs in their semantics, including non-exocentric uses of epistemic modals⁷ and sentential modifiers like *in X's opinion* or *presumably* (6).⁸ There seems to be an interesting gap. There are no doxastic expressions describing that an agent is in a position to make an inference regarding the truth of the prejacent based on some body of evidence, but doesn't do it (something like having reasons to believe without actually believing). I will simply state it here, without speculating why such expressions may be unattested despite being conceivable.

(5) DOXASTIC PREDICATES

#I **concluded/inferred/guessed/hoped/suspected/thought/was led to believe** that Maria was out of town, but I didn't believe it even then.

(6) OTHER FORMS OF MENTAL ATTITUDE REPORTS

- a. #It is **presumably** raining on Ferenginar, but I don't believe it.
- b. #**In Jake's opinion**, it is raining on Ferenginar, but he doesn't believe it.
- c. #It **must/might** be raining on Ferenginar, but I don't believe it.

To sum up, we see the same pattern as we have observed in the evidential domain: speech reports at large, including with hearsay evidentials, may allow non-endorsement, while mental attitude reports, including with non-hearsay evidentials, do not.⁹ It is therefore desirable to have a theory that derives (non-)endorsement with evidentials as a species of a more general

6. <https://bit.ly/3IkddxJ>

7. The term 'exocentric' refers to readings of perspectival expressions that do not in any way include the speaker, and comes from the literature on predicates of personal taste (Lasnik 2005 and later work). It has been contested whether epistemic modals actually have truly exocentric uses. What I am after is the following, uncontested observation. Whenever expressions like *must* and *might* are used in a way that the speaker is included in the relevant group whose knowledge is being talked about (DeRose 1991; von Stechow and Gillies 2010; Weatherson and Egan 2011), a modal claim is generally incompatible with complete suspension of belief, though it may sometimes be compatible with different shades of doubt. See von Stechow and Gillies (2021); Lassiter (2016) for further discussion.

8. In general, *presumably* behaves very much like inferential evidentials (Krawczyk 2012), while *in DP* is a shortcut for a family of at-issue intensional operators (von Stechow and Heim 2011; Smeijne 2024). English also has expressions of the form *given X*, which have a more involved semantics and talk about facts, rather than mental states (Kaufmann and Kaufmann 2020).

9. Note that doxastic predicates can talk about belief change and are therefore compatible with follow-ups *I don't believe it anymore*. Evidentials, even those that semantically make reference to the time of evidence acquisition (see discussion and references in Hirayama and Matthewson 2022; Johnson 2022), don't have separate temporal anchoring for the speaker's belief and are therefore always evaluated with respect to the time of utterance at the matrix level (I am not aware of any studies that investigate temporal anchoring of embedded evidentials). This difference does not affect my central claim: doxastic expressions require *some* credence in the prejacent at the index with respect to which they are evaluated.

phenomenon. It's worth emphasizing that the language of mental attitudes includes, but isn't limited to, epistemic modals, so treating non-hearsay evidentials as a species of attitude report does not necessarily commit us to anything with respect to the relationship between evidentiality and (epistemic) modality.

2.2 Lexical variation

The existing research focuses primarily on the mere possibility of non-endorsement. Thus, [AnderBois \(2014\)](#) argues that non-endorsement depends solely on context and that rich, evaluative language is crucial in enabling disavowals. As I demonstrate below, it is in fact worth dwelling on the linguistic shapes of disavowals, since they help us uncover a genuine case of lexical variation within hearsay overlooked in previous literature. Below I compare several hearsay expressions: English *apparently*, German *sollen*, Dutch *schijnen*, Turkish *mİş* and Japanese *sooda*. I want to stress that there is considerable inter-speaker variation, and in some cases, I suspect that we might be dealing with ongoing language change. Throughout the paper, I report the judgments of my consultants, but I believe that this area is ripe for an experimental investigation. Importantly, this variation further underscores my point: hearsay evidentials aren't created equal.

The most common diagnostic of non-endorsement is the availability of follow-ups of the form 'I don't believe that ϕ ' (the majority of cases in [AnderBois's \(2014\)](#) sample feature a version of this continuation). However, this does not necessarily indicate complete lack of endorsement, despite what was previously assumed. First, in English and many other languages, *believe* is a neg-raising predicate ([Gajewski 2007](#)): negation in the matrix clause is frequently, but not always, interpreted as negation of the complement. Thus, the statement 'I don't believe that ϕ ' is ambiguous between two interpretations: (i) **weak**: agnosticism about ϕ , and (ii) **strong**: belief that not ϕ . To the best of my knowledge, this issue remained under the radar in the discussion of non-endorsement with hearsay evidentials and it has not been systematically tested which interpretation the disavowal is picking out.¹⁰ Second, only some evidentials allow genuine non-endorsement with follow-ups of the form ' ϕ is not the case', a fact rarely discussed in the literature. I propose that disavowals allow us to gauge the speaker's attitude towards the prejacent and that evidentials fall into four (and possibly more) classes based on how much of this information is lexically encoded: lack of attitude, lack of knowledge, unsecludedness, and belief. On a methodological note, it is also important to pay attention to the verb. In English, *believe* is often judged as weaker than *think* ([Koev 2019, 2025](#)) but other languages are less studied in this respect; see also discussion before example (12) on the difference between 'don't believe' and 'can't believe'. With these caveats out the way, we can proceed.

Lack of attitude Some hearsay evidentials are compatible with the speaker's attitude of any kind, from fully endorsing the prejacent to explicit assertions that the prejacent is false. It is illustrated below for German (data of this sort have been extensively discussed in the literature; see, e.g., [Mortelmans 2000](#); [Schenner 2010b](#)) and Turkish, and similar data have been reported, for example, for Tagalog *daw* ([Kierstead 2015:43, ex.15b](#)), Cuzco Quechua *=si* ([Faller 2002:191, ex.152](#)) and Cheyenne *-séstse* ([Murray 2010:103, ex.4.5](#)). These evidentials seem to be closest to speech reports with *say*, which do not encode any information about the speaker's evaluation of the likelihood of the prejacent.

10. For example, [Smirnova \(2012:36, fn.12\)](#) argues that Bulgarian indirect evidential is compatible with the lack of belief because it can embed under *Ne vjarvam* 'I don't believe', but does not in fact show whether it's the strong neg-raised reading or the weak unopinionated one.

(7) German

Maria **soll** gestern verreist gewesen sein
 Maria **REP.3SG.PRES** yesterday be.away.PTCP be.PTCP be.INF
 ‘Maria **is supposed to** have been away yesterday ...’

- a. ...aber das ist nicht wahr.
 but that be.SG.PRES NEG true
 ‘...but this is not true.’
- b. ...aber ich glaube, dass das nicht stimmt.
 but I believe.1SG.PRES comp that NEG be.true.3SG.PRES
 ‘...but I believe it’s false.’
- c. ...aber das glaube ich nicht.
 but that believe.1SG.PRES I NEG
 ‘but I don’t believe it.’

(8) Turkish

Maria dün şehir dışında-y-**mış**
 Maria yesterday town outside-COP-**IND**
 ‘Maria was out of town yesterday (**I heard**) ...’

- a. ...ama bu doğru değil.
 but this true NEG.COP
 ‘...but this is not true.’
- b. ...ama bun-un yalan ol-duğ-un-u bil-iyor-um.
 but this-GEN false cop-NMLZ-3-ACC believe-PRES-1
 ‘...but I believe it’s false.’
- c. ...ama ben bun-a inan-mı-yor-um.
 but I this-DAT believe-NEG-PRES-1
 ‘...but I don’t believe it.’

Unsettledness of belief Some hearsay evidentials indicate that the speaker’s doxastic state may be unsettled with respect to the prejacent. They allow weak disavowals ‘I don’t believe ϕ ’, but are not compatible with stronger claims. This is illustrated below for English *apparently*¹¹ and Dutch *schijnen*. Similar data have been first discussed for Dutch by Koring (2013), who shows that *schijnen* does not allow flat-out contradictions and takes it to be evidence of its evaluative nature. AnderBois (2014) takes such data to indicate that non-endorsement is made easier with rich contexts, however, such view overlooks the categorical differences in the

11. Matrix-clause *apparently* can be compatible with the speaker’s belief that $\neg\phi$ in the following cases:

- (i) I was talking to my students. Apparently, I change the class schedule on a whim without ever updating the syllabus. I will need to set the record straight.

This is an instance of Free Indirect Discourse, a special literary technique for representing speech and thoughts of a protagonist (Eckardt 2014). Even though no overt clause-taking verb may be present, Free Indirect Discourse is usually analyzed as involving a covert attitudinal operator (Doron 1991; Sharvit 2008), which in turn may interact with the interpretation of evidential expressions in its scope, much like overt embedding predicates. Evidentiality and Free Indirect Discourse is uncharted territory well beyond the scope of this paper. Of importance here is that such cases should not be treated on a par with unembedded cases discussed in the main text. I suspect that some of the examples used by Smirnova (2012) to demonstrate that Bulgarian evidential does not encode belief (p.4, ex.7) fall under the same umbrella and therefore should not be considered instances of non-endorsement. In other contexts the Bulgarian evidential behaves more like *apparently* (compatible with agnosticism, but not full endorsement that $\neg\phi$), rather than German *sollen* (no attitude).

attitudes signalled by different continuations. As I argue extensively in Section 3.3, these data are important in understanding the nature of the attitude encoded by different evidentials.

(9) **Apparently**, Maria was out of town ...

- a. ...✓but I don't think it's true.
- b. ...# but I believe that this isn't the case.
- c. ...# but I know it's not true/but I know she wasn't.

(10) Dutch

Maria **schijnt** weg te zijn...

Maria **REP.3SG.PRES** away to be.INF

'Maria was away (**I heard**) but ...'

- a. ...maar dat geloef ik niet.
but that believe.1SG.PRES I NEG
'...but I don't believe it.'
- b. #...maar ik denk dat ze in de stad was.
but I think.1SG.PRES COMP she in DEF town be.3SG.PST
Intended: '...but I think that she was in town.'
- c. #...maar ik weet dat ze hier was. Ik heb
but I know.1SG.PRES COMP she here be.3SG.PST I have.1SG.PRES
haar gezien!
her see.PTCP
Intended: '...but I know that she was here. I've seen her!'

Unsettledness of knowledge Belief doesn't always supply sufficient grounds for assertion. Thus, there is a subtle difference between being in a position to say that one doesn't believe ϕ vs. saying that ϕ is not the case. (11) illustrates.

(11) I believe that it is not raining on Ferenginar, but I don't know for sure.

I argue that some hearsay evidentials make reference to this distinction in their semantics and encode that the speaker isn't certain whether their belief about the prejacent is true. These evidentials allow disavowals of the form 'I believe not ϕ ', but not ' ϕ is not true' or 'I know not ϕ '. I illustrate it below for Japanese *sooda*.¹² Initial research suggested that *sooda* is a generic speech report compatible with assertions of falsehood (McCready and Ogata 2007:161-162), a claim refuted by Matsubara's (2017) experimental findings (pp.73-110). In particular, utterances with *sooda*, unlike with overt speech verbs, are judged as contradictory when followed up by 'but not ϕ ' regardless of the reliability of report. However, Matsubara does not discuss disavowals with 'believe' and, as I discuss in Section 3.3, predicts that all disavowals will be impossible. Moreover, some of McCready and Ogata's (2007) examples contain 'can't believe' and therefore might not be true disavowals. In English and several other languages 'can't believe'—but not 'don't believe'—is a veridical marker of surprise about the prejacent (Roberts 2019), and thus conveys endorsement despite earlier belief that $\neg\phi$. Such data only underscore my earlier point that more language-specific research is needed on the exact shape of disavowals.

12. As discussed in detail by McCready and Ogata (2007), *sooda* has two evidential uses with different syntax and semantics. All my examples feature the hearsay *sooda*, which takes tensed complements.

(12) Japanese

Maria-wa Bahama-ni shunookeringu-ni it-ta **sooda**
 Maria-TOP Bahaamas-to snorkeling-to go-PST **REP**
 ‘Maria went snorkeling in the Bahamas (**I heard**) ...’

- a. #...ga it-tei-nai.
 but go-RES-NEG
 ‘...but she didn’t go.’
- b. ✓it-tei-nai-to omou
 go-RES-NEG-COMP think
 ‘...but I think that she didn’t.’
- c. ✓...ga sore-ga jijitsu-da-to-wa omowa-nai
 but it-NOM fact-COP-COMP-TOP think-NEG
 ‘...but I don’t think that it’s the case.’
- d. ✓it-tei-nai daroo
 go-RES-NEG must
 ‘...but she must not have gone.’

Belief Hearsay evidentials in several languages spoken in British Columbia (Canada) have been described as incompatible with suspension of belief: Gitksan (Tsimshianic; [Peterson 2010](#)), Nuu-chah-nulth (Wakashan; [Waldie 2013](#)) and St’át’imcets (Salish; [Matthewson et al. 2007](#)). [AnderBois \(2014\)](#) suggests that these languages simply lack mechanisms for pragmatic shift, since they present an unexplained exception to an otherwise robust pattern. First, as discussed above, specific evidentials do vary in which disavowals they allow and only some of them indicate complete lack of endorsement, which is in conflict with AnderBois’ view. Second, while it would be interesting to conduct more detailed fieldwork on these languages, the existence of hearsay evidentials that ban disavowals of all stripes is only natural if we take the parallels between hearsay evidentiality and communicative predicates more seriously.

Recent years have seen a steady interest in the cross-linguistic semantics of clause-taking predicates. Compared to doxastics (*believe*, *know*) or bouletics (*want*, *wish*), the fine-grained semantics of communicative predicates received less attention (though see [Anand, Grimshaw, and Hacquard 2017](#); [Anand and Hacquard 2014](#); [Grimshaw 2015](#); [Korotkova and Anand 2025](#); [Rudin 2025b](#)). The literature on evidentiality implicitly assumes that communicative predicates always allow non-endorsement when used non-parenthetically. This assumption is incorrect: the availability and nature of (non-)endorsement varies across predicates and constructions. English, for example, has veridical communicative predicates (or rather, predicates compatible with communicative uses) *be right*, *confirm* or *demonstrate*.¹³ As (13) illustrates, these predicates entail endorsement (more precisely, the prejacent’s truth at a higher world), unlike *state* or *claim* (3b), which do not semantically encode such information.

- (13) a. Ana **is right** that Maria is out of town ...
 ... #but she never said it. [voiding report]
 ... #but that’s not true/#but I don’t believe it. [voiding endorsement]
- b. The papers **confirmed/demonstrated/established** that there is a new trade agreement with the Dominion ...

13. These predicates have other, non-communicative uses ([van Elswyk 2025](#)), but the same is true of many evidentials. My present concern is their uses in speech reports.

...#but they never reported on this.	[voiding report]
...#but that's not true/#but I don't believe it.	[voiding endorsement]

Those hearsay evidentials that allow disavowals vary in the exact type of continuation they are compatible with, which is a strong argument against proposals that advocate a one-size-fits-all approach to non-endorsement. Thus, [AnderBois \(2014\)](#) argues that non-endorsement is due to a pragmatic mechanism available to all hearsay evidentials (and possibly speech reports in general, though this is not discussed) and explicitly claims that there is no lexical variation provided that the language in question has relevant mechanisms that would enable the pragmatic shift to the original author of the report. This view incorrectly predicts that all hearsay evidentials behave the same way. However, as revealed by the shape of follow-ups that they license, some hearsay evidentials actually do encode the information about the speaker's attitude. I propose a more nuanced picture: non-endorsement with hearsay follows a more general pattern found in the domain of speech reports and is subject to lexical variation.

Natural language has speech reports that convey non-detachable information about the speaker's attitude towards the prejacent. We do not know how common such expressions are. English, for example, has great propensity for expressing information via clause-taking predicates, but subtle lexical—rather than structural—distinctions within such predicates are still very much understudied even on the material of familiar and accessible languages. Furthermore, it could be the case that languages with rich evidential systems use them to express the notions that would be conveyed via other means elsewhere, including via communicative predicates. And if we have veridical communicative predicates, we might as well have hearsay evidentials that lexically encode endorsement. I hypothesize, though cannot prove anything at the moment without further fieldwork, that hearsay evidentials in Gitksan, Nuu-chah-nulth and St'át'imcets are precisely of that nature. We should therefore come up with an analysis of the non-endorsement puzzle that doesn't preclude their existence outright on conceptual grounds.

On a more general note, the space of lexical variation within hearsay evidentials is a rich and underexplored area. There is definitely some degree of variation, e.g., in whether secondhand reports are allowed ([Willett 1988](#)), or whether sources can be expressed overtly ([Krawczyk 2012](#)). That said, much research within formal semantics primarily aims at parameterizing the behavior of hearsay evidentials using binary features like reliability ([Krawczyk 2012:277-294](#)), often reducing the variation to the putative modal vs. non-modal distinction ([Faller 2002, 2019](#); [Matthewson et al. 2007](#)) and treating hearsay evidentiality as a monolithic construction in the taxonomy of speech reports ([Bary and Maier 2021](#)). At the same time, given how many shades of meaning communicative predicates can express, it would be interesting to see to what extent any of these distinctions play a role in the meaning of evidential expressions. For example, it is known that communicative predicates differ in their selectional restrictions: not all of them take quotations as complements ([Grimshaw 2015](#)), only some of them take interrogative complements ([Rudin 2025b](#)), and only some embed imperatives ([Kaufmann and Poschmann 2013](#); [Kaufmann 2014](#)). Some hearsay evidentials can also be combined with non-declarative prejacent, e.g., Tagalog *daw* ([Schwager 2010](#); [Thomas 2014](#)) or Cuzco Quechua *=si* ([Faller 2002](#)). This property has often been taken as evidence of a special status of those evidentials that allow them compared to those that don't ([AnderBois 2017](#); [Faller 2007, 2019](#)), but I suspect that it can be modelled using the general mechanisms of complement selection, in line with communicative predicates.

2.3 Recap

The divide between the language describing reported speech vs. mental attitudes cross-cuts the grammar. For example, communicative predicates (*assert*, *claim*, *posit*) differ from representational attitude predicates (*discover*, *infer*, *think*) along several syntactic and semantic dimensions, including argument structure, factivity or what they license in their complements. I propose that the non-endorsement puzzle is best understood through the prism of this general distinction, and that any theory of non-endorsement should aim at explaining the following facts in a principled way:

- mental attitude reports, including non-hearsay evidentials, require endorsement;
- speech reports, including hearsay evidentials, may allow non-endorsement;
- speech reports, including hearsay evidentials, are subject to lexical variation in the availability of non-endorsement and the shape of disavowals.

The next section provides an intensional account that captures these desiderata, though I want to stress that this is not the only analytical option.

3 Informational modality redux

Approaches to evidential meaning can be grouped into two families (see extensive overviews in [Korotkova 2026](#); [Murray 2020, 2021](#)).¹⁴ According to one light of thought (‘the modal view’), evidentials are best treated as intensional operators within possible world semantics ([Izvorski 1997](#); [Matthewson et al. 2007](#); [Schwager 2010](#); [Schenner 2010b](#) a.o.). According to another line of thought (‘the illocutionary view’), evidentials are best treated as operators directly interacting with the structure of discourse within dynamic frameworks ([Davis et al. 2007](#); [Faller 2019](#); [Murray 2014](#); [Northrup 2014](#) a.o.). This isn’t a purely modelling dispute: it has often been argued that different evidential expressions fall into (at least) two semantic categories, modal vs. non-modal, with a corresponding analysis as intensional vs. extensional operators (see [Korotkova 2016b, 2021](#) for empirical arguments against the semantic heterogeneity view). This literature implicitly assumes the following (with the notable exception of [McCready and Ogata 2007](#)): that (i) there are two distinct categories of expressions, (epistemic) modals and update operators; and that (ii) bona fide modals (e.g., *must* and *might*) are best analyzed as propositional operators, as it is done within the Kratzerian tradition ([Kratzer 1977, 1981, 1991, 2012](#)). However, it’s just but one view on modality in natural language—for example, dynamic frameworks shift the focus away from the truth conditions of epistemic modals to their update potential ([Veltman 1996](#); [Willer 2013](#); [Yalcin 2011](#), see overview in [Willer 2015](#)), so it’s important to come up with diagnostics of modality that are not directly tied to discourse dynamics.

Endorsement and non-endorsement have also been extensively discussed in the literature as possible diagnostics of modality ([Faller 2002](#); [Papafragou 2006](#)). Thus, it has been often argued that the availability of follow-ups that explicitly state that the prejacent of an evidential claim is true or false speaks against a modal analysis, since modal expressions are often taken to encode uncertainty. In what follows, I argue that the availability of different continuations targeting (non-)endorsement diagnoses whether the expressions in question encode that the speaker’s information state is unsettled with respect to the prejacent, which is in and of

14. For the purposes of this paper, I put aside pragmatic approaches, which derive evidential grounding as a side-effect of event structure (see discussion in [Korotkova 2026](#)).

itself independent of modality. I will understand modality broadly as a semantic category encompassing a host of expressions that make reference to intensional domains (cf. [Vetter 2011](#)), including modal auxiliaries and attitude verbs. Under this view, both communicative predicates (which often encode no information about beliefs) and factive and veridical predicates (which encode that the information state is settled about the prejacent) are modal expressions. Importantly, this characterization allows us to focus on the formal properties of expressions at hand. I start by showing that several hearsay evidentials, much like communicative verbs, give rise to intensional effects: they allow contradictions in their scope and they can block presuppositions. I then address a misconception concerning a version of the modal view and, building on the literature on speech reports, delineate an informational modal blueprint that can account for the lexical variation discussed in Section 2.2 by incorporating information about the speaker’s knowledge and belief into the semantics of respective expressions. My overarching goal is to vindicate a modal analysis for at least some hearsay evidentials, in hopes that empirical diagnostics that I offer will be used in later research.

3.1 Intensional domains

Debates around the modal vs. non-modal nature of evidentiality often focus on the (dis-)similarities between evidentials and specific modals, such as English *must* and *might*, or their counterparts in other languages. While this line of research proved fruitful for understanding the behavior of evidentials in a variety of languages, it has led to a narrow focus on a subclass of modal expressions and prevented us from examining evidentials against a larger context of intensional environments. I redress this gap by focusing on two empirical diagnostics that allow us to detect evidential intensionality: the possibility of contradictions and presupposition blocking. In each case, evidentials pattern like communicative predicates (a canonical intensional environment), which in turn warrants an analysis of those evidentials as intensional operators.¹⁵

Contradictions Intensional environments, including speech and attitude reports, are known to ameliorate contradictions because they allow different parts of the prejacent to be evaluated with respect to different worlds: expressions in the main predicate position must be read ‘de dicto’ while expressions outside of main predicate position can be read ‘de re’ or ‘de dicto’ ([Anand and Korotkova 2022](#); [Farkas 1997](#); [Keshet 2010](#); [Musan 1997](#); [Schwarz 2012](#)). Consider (14).¹⁶

(14) Contradictory prejacent

You read an article about a very dangerous disease, claimed to be caused by a fungus, while in fact you know that it is a virus.

- a. #The deadliest virus is not a virus.
- b. **Apparently**, the deadliest virus is not a virus.

15. One of the evidentials I discuss is German *sollen*. The literature doesn’t doubt that it’s a modal semantically because it belongs to the syntactic category of modal verbs. My point is that we need to establish its intensionality independently by looking at semantic diagnostics.

16. Note that the Dutch example (14f) features a discourse particle, *helemaal* ‘totally/at all’. However, unlike English *actually*, it doesn’t seem to improve contradictions on its own and the corresponding sentence without *schijnen* is infelicitous.

- c. German
 Der tödlichste Virus **soll** kein Virus sein.
 DEF deadly.SUP virus.NOM.SG **REP** NEG.INDEF virus.NOM.SG be.INF
 ≈ ‘**I hear** that the world’s deadliest virus is not a virus’.
- d. Turkish
 Dünya-nın en ölümcül virüs-ü virüs değil-**miş**.
 world-GEN SUP deadly virus-ACC virus NEG.COP-**IND**
- e. Japanese
 Ano-kikenna-uirusu-wa uirusu-de-wa-nai-**sooda**.
 that-dangerous-virus-TOP virus-COP-TOP-NEG-**REP**
- f. Dutch
 Het dodelijkste virus **schijnt** helemaal geen virus te zijn.
 def deadly.SUP virus **REP.3SG.PRES** PTCL NEG.INDEF virus to be.INF

In (14), a range of evidentials allow contradictions in their scope. Sometimes, as in Japanese, the acceptability of the sentence is improved by a special intonation that signals irony or sarcasm with respect to a faulty claim. Crucially, none of those sentences are felicitous on their own.¹⁷ In this respect, hearsay evidentials pattern like communicative predicates and other intensional speech reports, such as English *according to* (15).¹⁸ Note also that in each case, the speech report is not an instance of direct discourse—quotations are opaque and do not allow ‘de re’ readings, thus leading to attributing a contradictory claim to the original author of the report, as in (15b).

- (15) a. The newspaper claimed that the deadliest virus is not a virus.
 b. #The journalists reported: “The deadliest virus is not a virus.”
 c. According to the newspaper, the deadliest virus is not a virus.
 d. German
 Der Zeitungsartikel behauptet, dass der tödlichste Virus kein
 DEF newspaper.article claim.3SG.PRES COMP DEF deadly.SUP virus NEG.INDEF
 Virus ist.
 virus.NOM.SG be.3SG.PRES
 ‘The newspaper article claims that the deadliest virus is not a virus.’
 e. Turkish
 Gazete en ölümcül virüs-ün bir virüs ol-ma-dığ-ın-ı iddia
 newspaper SUP deadly virus-GEN INDEF virus COP-NEG-NMLZ-3-ACC claim
 ed-iyor
 make-PRES
 ‘The newspaper claims that the deadliest virus is not a virus.’

17. The English sentence (a) may be non-contradictory with a special quotative intonation (cf. [Stec et al. 2016](#)) or with scare quotes. In each case, it will be an instance of mixed quotation ([Maier 2014](#)), an intensional environment; cf. discussion in ([Anand and Korotkova 2022:623, fn.4](#)).

18. The German sentence (15d) is further improved by using the reporative subjunctive, often used in news reporting and contributing to the general distancing effect ([Fabricius-Hansen and Sæbø 2004](#)), but this isn’t obligatory.

f. Japanese

Ronbun-wa ano-kikenna-uirusu-wa uirusu-de-wa-nai-to shuchoosi-tei-ru
 paper-TOP that-dangerous-virus-TOP virus-COP-TOP-NEG-COMP claim-PROG-PRES
 ‘The newspaper claims that that dangerous virus is not a virus.’

g. Dutch

De krant meldt dat het dodelijkste virus geen virus
 DEF newspaper report.3SG.PRES COMP DEF deadly.SUP virus NEG.INDEF virus
 is
 be.3SG.PRES
 ‘The newspaper reports that the deadliest virus is not a virus.’

Following the literature on clause-taking predicates, I will treat both (14) and (15) as an instance of a ‘de re’ reading of the expression in the subject position and thus evidence of evidential intensionality, since all accounts of ‘de re’ in one way or another involve intensional quantification.

Presupposition blocking Another diagnostic that I propose for evaluating evidential intensionality is presupposition blocking. In a sense, it is a mirror image of modal subordination, which has been previously used in the literature on evidentiality (see discussion and references in [Korotkova 2026](#)). A variety of environments, including speech and attitude reports, can block the presuppositions of their prejacent.¹⁹ Below I show that at least some hearsay evidentials pattern like communicative predicates and don’t require the presuppositions of their prejacent to project to the matrix level. Consider (16).²⁰

(16) **Presupposition blocking**

You read in a reputable newspaper that Switzerland stopped drilling in the Arctic.

a. **Apparently**, Switzerland stopped drilling in the Arctic, though actually I don’t think they ever drilled there.

b. German

Die Schweiz **soll** in der Arktis nicht mehr bohren. Aber hä?
 DEF Sw. **REP.3SG.PRES** in DEF Arctic NEG more drill.INF but PRCL
 Die Schweiz hat doch gar nicht in der Arktis gebohrt.
 DEF Sw have.3SG.PRES PTCL PTCL NEG in DEF arctic drill.PRPC
 ‘Switzerland supposedly has stopped drilling in the Arctic. But wait a minute, Switzerland hasn’t drilled there at all.’

c. Turkish

İsviçre Arktik’te sondaj yapmay-ı durdur-**muş** ama orda sondaj
 Switzerland Arctic.IN drill making-ACC stop-**IND** but in.there drill
 yap-tık-lar-in-i bile düşün-mü-yor-um
 make-NMLZ-PL-3.POSS-ACC even believe-NEG-PRES-1
 ‘I heard that Switzerland stopped drilling in the Arctic, but I don’t believe they even drilled there.’

19. Epistemic modals like *must* and *might* are presupposition holes, so this is a case in which at least some modals and some evidentials diverge. I believe that this has to do with the fact that epistemics require the prejacent to be a live possibility (cf. also footnote 7), but won’t elaborate on it further.

20. Note that in the Dutch example (16d) the preceding clause is simply specifying the source; its equivalent with clausal embedding (*schijnen* under ‘read’) is infelicitous, unlike clausal embedding without the evidential.

d. Dutch

Ik las net in het NRC. Zwitserland **schijnt** gestopt te zijn
 I read.PST just in DEF NRC Switzerland **REP.3SG.PRES** stop.PTCP to be.INF
 met boren op de Noordpool, maar ik denk niet dat ze
 with drill.INF in DEF North.pole but I think.1SG.PRES NEG COMP they
 daar Æijberhaupt hebben geboord.
 at.all have.3PL.PRES drill.PTCP
 ‘I just read in the NRC [Dutch newspaper]. Switzerland has apparently stopped
 drilling on the North Pole, but I don’t think they drilled there at all.’

As (16) shows, English *apparently*, German *sollen*, Dutch *schijnen* and Turkish *mİş* can block the presupposition of the aspectual verb ‘finish’ (across languages, this presupposition often behaves as a soft trigger and is therefore relatively easy to cancel). All sentences above are infelicitous without the evidential, as they attribute contradictory beliefs to the speaker. Note that not all hearsay evidentials easily allow this reading—for example, Japanese *sooda* is judged as infelicitous in discourses above. I suspect that the availability of the non-projected reading correlates with the degree of belief those evidentials are associated with, though I won’t explore it in detail here. Importantly, those evidentials that do allow presupposition blocking pattern exactly like communicative predicates (17).

(17) a. I’ve read in a newspaper that Switzerland stopped drilling in the Arctic, but I don’t think it ever drilled there before.

b. German

Ich habe in der Zeitung gelesen, dass die Schweiz in der
 I have.1SG.PRES in DEF newspaper read.PTCP COMP DEF Sw. in DEF
 Arktis nicht mehr bohrt. Aber hä? Die Schweiz hat dcht
 Arctic NEG more drill but PRCL DEF Sw have.3SG.PRES PTCL
 gar nicht dort gebohrt.
 PTCL NEG there drill.PTCP
 ‘I’ve read in the newspaper that Switzerland stopped drilling in the Arctic. But wait
 a minute, Switzerland has never drilled there.’

c. Turkish

Gazete İsviçre’nin Arktik’te sondaj yapmay-ı durdur-duğ-un-u
 newspaper Switzerland.GEN Arctic.IN drill making-ACC stop-NMLZ-3-ACC
 yaz-dı ama orda sondaj yap-tık-lar-ın-ı bile düşün-mü-yor-um.
 write-PST but in.there drill make-NMLZ-PL-3-ACC even believe-NEG-PRES-1
 ‘A/the newspaper wrote that Switzerland stopped drilling in the Arctic but I don’t
 believe they even drilled there.’

To sum up, the novel data on contradictions and presupposition blocking demonstrate that the evidential operator semantically interacts with its prejacent and introduces an intensional domain, behaving much in the same way as communicative predicates. I take such data to support the modal view on evidentiality at least for those constructions. Interestingly, [Faller \(2002:256-257\)](#) briefly mentions that Cuzco Quechua *=si* allows local accommodation of a presupposition introduced by the prejacent, but maintains that *=si* is not an intensional operator and does not bring up such data in later work. I strongly suspect that *=si* can be treated along the lines I propose below and that it simply differs from more canonical modals.

3.2 Refining arguments against the epistemic modal account

In the very first formal-semantic work on evidentiality (Izvorski 1997), Pancheva proposed that Bulgarian evidential perfect can be treated as an epistemic modal within the Kratzerian framework. Roughly speaking, it is analyzed as a universal quantifier over worlds associated with an epistemic modal base (which provides worlds compatible with what is known to the speaker) and a stereotypical ordering source (which ranks the worlds in the modal base according to their normality), in parallel to Kratzer’s original proposal for epistemic modals modulo the presence of evidential restrictions and similar to a later proposal for *must* (von Fintel and Gillies 2010, 2021). An evidential utterance will be true just in case the prejacent is true in the most normal of the epistemically-accessible worlds. This account straightforwardly captures the intensionality of at least some hearsay evidentials discussed in the previous section, but faces problems of its own.

An oft-cited argument against this view runs as follows (Faller 2002, 2019; Smirnova 2012, a.o.). If evidentials are modelled as epistemic modals, we expect them to pattern like prototypical epistemic modals *must* and *might*. However, as is well-known, epistemic modals are incompatible with disavowals in non-exocentric interpretations (6c, repeated below as 18). Therefore, the epistemic modal account must be wrong: it *undergenerates* for hearsay evidentials, incorrectly predicting the impossibility of disavowals.

(18) #It **must/might** be raining on Ferenginar, but I don’t believe it. [=6c]

This reasoning is flawed. Rehashing Yalcin’s (2007) argument against Kratzer’s original account of epistemics, we can show that the epistemic account doesn’t work for the Bulgarian evidential and similar expressions, but for a different reason. Bulgarian evidential perfect is compatible with inference and hearsay (and possibly other sources; Smirnova 2012). However, in Bulgarian and elsewhere, only hearsay allows non-endorsement. Pancheva’s account does not hard-wire endorsement and predicts that the actual world needn’t be part of the worlds selected by the ordering source, which in turn opens the door for disavowals (though the paper doesn’t discuss disavowals explicitly, it mentions rumors; Izvorski 1997:231). This is a welcome result for hearsay, but the system in fact *overgenerates* for inferential readings, since nothing prevents the speaker from disavowing such claims as well (this is precisely Yalcin’s point about *might* and *must* in Kratzer’s system). To conclude, the epistemic modal account indeed makes incorrect predictions for evidentials, because it doesn’t capture the robust difference between hearsay vs. non-hearsay evidentials in the availability of non-endorsement, one of the desiderata outlined in Section 2.

3.3 Refining the informational modal account

My main goal in this section is to provide a blueprint for hearsay evidentials as intensional operators in a way that captures both the principled differences between hearsay vs. non-hearsay and the lexical variation within hearsay. I will use the a version of Hintikka semantics for speech reports (Bary and Maier 2021; Kaufmann and Kaufmann 2020), though nothing hinges on this choice and my primary focus is the conceptual structure of the analysis. The term ‘informational modality’ comes from Kratzer (2012) and I use it to highlight that hearsay evidentials can be treated as modals, albeit of a different kind than epistemics.

I propose that for all hearsay evidentials (including hearsay uses of indirect evidentials) the domain of quantification is the contents of reports available to the speaker at the world of evaluation. Non-hearsay evidentials (including inferential uses of indirect evidentials), provided

that they should be analyzed as intensional operators at all, can be treated as having different quantification domains, for example, beliefs. This modelling choice tracks the fundamental distinction between hearsay vs. non-hearsay evidentiality, which, in turn, is a reflection of a much broader distinction between speech vs. attitude reports. As discussed in Section 2.1, expressions that deal with reported information allow one not to believe the prejacent and expressions that deal with mental attitudes require one to consider the prejacent a live possibility. By making a distinction in the quantificational domains for different evidentials, we thus fulfill one important desideratum for a theory of non-endorsement: only hearsay evidentials allow it. One issue that I will not discuss here is co-lexification: for many expressions discussed here hearsay is not the only reading. Thus, German *sollen* has a life as a weak necessity deontic modal, and English *apparently* or Turkish *mlş* are also used in non-hearsay abductive inference contexts (2). Modal expressions, including modal auxiliaries and attitude predicates, are known to be compatible with different modal flavors, so this could be used as an additional argument in favor of assimilating evidentials and modals, but I will not pursue it further.

I further propose that the variation within hearsay evidentials with respect to endorsement can be also captured within this general framework by adding constraints on the speaker’s information state. Previous literature has focused on the categorical contrast between evidentials that allow the speaker to disavow the prejacent vs. those that don’t. Thus, it has often been argued that the possibility of non-endorsement reflects the difference between modal vs. non-modal evidentials (see also discussion in Section 4). And Kratzer (2012:34-35) argues that evidentials that require endorsement are associated with realistic, and not informational, conversational backgrounds (cf. also Kaufmann and Kaufmann 2020). However, as we have seen in Section 2.2, the shape of disavowals different hearsay evidentials are compatible with demonstrates that the actual picture is more nuanced and that there are more than two types of hearsay evidentials. I propose that this variation stems from the fact that some hearsay evidentials, just like some communicative predicates, additionally give rise to inferences about the speaker’s attitude towards the prejacent and that these inferences come in several varieties: lack of attitude, lack of knowledge, unsettledness, and presence of belief. I parameterize those differences below as at-issue entailments about the speaker’s doxastic state, in parallel to the account of the veridicality of some communicative predicates (Anand and Hacquard 2014). Given that evidentials almost always take wide-scope with respect to negation (see references and discussion in Korotkova 2020) and exhibit a peculiar behavior in other downward-entailing environments, I will not further elaborate on the exact status of such inferences beyond the fact that they are part of the conventional meaning of respective expressions.

Hearsay evidentials may vary in their selectional restrictions, sources they are compatible with or the possibility to express the source overtly. I concentrate exclusively on the variation in the attitude they express and put those other differences aside.

Lack of attitude Some hearsay evidentials allow disavowals of all stripes, including follow-ups asserting that the prejacent is false. I propose that such evidentials have the same semantics as *according to* and quantify over the worlds compatible with the contents of a report available to the speaker at the world of evaluation. This minimal semantics does not, by design, capture the discourse status of the evidential signal, and I come back to this issue in Section 5.1. Note that this semantics treats evidentials as always speaker-oriented. This is a simplification since I only discuss the behavior of evidentials at the root level, but it can be adjusted accordingly to account for the more complex patterns of behavior in attitudinal environments (Korotkova 2015, 2016b; Schenner 2010b,a; Schwager 2010).

- (19) a. Let $\text{CONTENT}(x, w)$ be the propositional content associated with the source x in w and R an acquaintance relation.
- b. $\llbracket \text{HEARSAY}_{\text{NO-ATT}} \phi \rrbracket^{c,g,w} = \exists x. \forall w' \in \text{CONTENT}(x, w). \llbracket \phi \rrbracket^{c,g,w'} \wedge \exists R. R(Sp, w, \text{CONTENT}(x, w)).$

The semantics in (19) is a notational variant of the standard semantics for many speech predicates and, modulo the presence of an overt source of report, adopts Kaufmann and Kaufmann’s (2020) analysis of *according to* adverbials. It only requires that the speaker is acquainted with some propositional content, which reflects the fact the many hearsay evidentials have little restrictions on the source. Just like Schenner’s (2010b) formal implementation of the same intuition for German *sollen*, this semantics predicts that the evidential operator shifts the world at the which the truth of the prejacent is evaluated, thus allowing for intensional effects, and does not say anything about the speaker’s attitude towards the prejacent, which captures strong disavowals. I illustrate it for the hearsay use of Turkish *mlş*.

- (20) a. Turkish
 Maria dün şehir dışında-y-**mlş** ama bu doğru değil. [=8a]
 Maria yesterday town outside-COP-**IND** but this true NEG.COP
 ‘I heard that Maria was out of town yesterday but that is not true.’
- b. $\llbracket (20a) \rrbracket^{c,g,w}$
 $= \exists x. \forall w' \in \text{CONTENT}(x, w). [\text{OUT.OF.TOWN}(Maria, w')]$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w))$
 $\wedge \neg [\text{OUT.OF.TOWN}(Maria, w)].$

(20a) is not a contradiction, which is successfully derived by the semantics in (19), as the truth of two complementary propositions is evaluated at different worlds. Any weaker claim that makes reference to the speaker’s belief about the prejacent is also acceptable, for the same reasons. Likewise, this semantics correctly predicts that follow-ups conveying different shades of endorsement will be allowed, including ‘I believe ϕ ’ and ‘ ϕ is actually true’. As mentioned in the introduction to this section, such follow-ups have been argued to be incompatible with a modal analysis, but this is only true if modality is taken to denote uncertainty. Under the view advocated here, such data do not present any problems.

- (21) a. Turkish
 Maria dün şehir dışında-y-**mlş**. Bu doğru.
 Maria yesterday town outside-COP-**IND** this true
 ‘Maria was out of town yesterday (I heard). It is true.’
- b. $\llbracket (21a) \rrbracket^{c,g,w}$
 $= \exists x. \forall w' \in \text{CONTENT}(x, w). [\text{OUT.OF.TOWN}(Maria, w')]$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w))$
 $\wedge [\text{OUT.OF.TOWN}(Maria, w)].$

Unsettledness of belief Some hearsay evidentials only allow weak disavowals of the form ‘I don’t believe ϕ ’, however, contra to what was previously assumed in the literature, these follow-ups do not signal complete non-endorsement, since these evidentials are incompatible with ‘I don’t believe ϕ ’ or ‘ ϕ is not the case’. I propose that such evidentials encode that the speaker’s doxastic state is unsettled with respect to the prejacent.

- (22) a. Let $\text{CONTENT}(x, w)$ be the propositional content associated with the source x in w , R an acquaintance relation and $\text{DOX}(Sp, w)$ the set of worlds compatible with the speaker's beliefs at w .
- b. $\llbracket \text{HEARSAY}_{\text{NO-BELIEF}} \phi \rrbracket^{c,g,w} = \exists x. \forall w' \in \text{CONTENT}(x, w). \llbracket \phi \rrbracket^{c,g,w'} \wedge \exists R. R(Sp, w, \text{CONTENT}(x, w)) \wedge \exists v \in \text{DOX}(Sp, w). [\phi(v)] \wedge \exists v \in \text{DOX}(Sp, w). [\neg \phi(v)].$

The semantics in (22) predicts that strong disavowals are banned. I illustrate it below for English *apparently*.

- (23) a. #Apparently, Maria was out of town, but I believe that it's not true/ that she wasn't.
[=9a]
- b. $\llbracket (23a) \rrbracket^{c,g,w} = \exists x. \forall w' \in \text{CONTENT}(x, w). [\text{OUT.OF.TOWN}(Maria, w')] \wedge \exists R. R(Sp, w, \text{CONTENT}(x, w)) \wedge \exists v \in \text{DOX}(Sp, w). [\text{OUT.OF.TOWN}(Maria, v)] \wedge \exists v' \in \text{DOX}(Sp, w). [\neg \text{OUT.OF.TOWN}(Maria, v')] \wedge \forall w'' \in \text{DOX}(Sp, w). [\neg \text{OUT.OF.TOWN}(Maria, w'')].$

The semantics in (22) correctly predicts that strong disavowals are contradictory with those hearsay evidentials that require the speaker to be undecided about the prejacent. On the one hand, the speaker considers both the prejacent and its complement a live possibility. On the other, the follow-up indicates that the speaker in fact believes the complement. A weaker follow-up, such as 'I don't believe ϕ ', is acceptable since it doesn't lead to a contradiction:

- (24) $\neg[\forall w' \in \text{DOX}(Sp, w). [\text{OUT.OF.TOWN}(Maria, w')]] = \exists w' \in \text{DOX}(Sp, w). [\neg \text{OUT.OF.TOWN}(Maria, w')]$

Note that this semantics also correctly predicts that follow-ups explicitly asserting the truth of the prejacent, or the speaker's belief in the prejacent, will be infelicitous, precisely because the speaker is undecided and the follow-up is in conflict with it.

- (25) a. #Apparently, Maria was out of town. I believe that this is true.
- b. $\llbracket (25a) \rrbracket^{c,g,w} = \exists x. \forall w' \in \text{CONTENT}(x, w). [\text{OUT.OF.TOWN}(Maria, w')] \wedge \exists R. R(Sp, w, \text{CONTENT}(x, w)) \wedge \exists v \in \text{DOX}(Sp, w). [\text{OUT.OF.TOWN}(Maria, v)] \wedge \exists v' \in \text{DOX}(Sp, w). [\neg \text{OUT.OF.TOWN}(Maria, v')] \wedge \forall w'' \in \text{DOX}(Sp, w). [\text{OUT.OF.TOWN}(Maria, w'')].$

Lack of knowledge Some hearsay evidentials allow the speaker to assert the lack of belief in the prejacent, but not to assert knowledge about an opposite claim. I propose that these evidentials track the gap between belief (that doesn't require justification) and knowledge (that does), though one can also model it by directly making reference to the quality of speaker's evidence, which is sufficient for belief and insufficient for knowledge and/or assertion.

- (26) a. Let $\text{CONTENT}(x, w)$ be the propositional content associated with the source x in w , R an acquaintance relation and $\text{EPI}(Sp, w)$ the set of worlds compatible with what the speaker's knows at w .
- b. $\llbracket \text{HEARSAY}_{\text{NO-KNOWLEDGE}} \phi \rrbracket^{c,g,w} = \exists x. \forall w' \in \text{CONTENT}(x, w). \llbracket \phi \rrbracket^{c,g,w'}$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w))$
 $\wedge \exists v \in \text{EPI}(Sp, w). [\phi(v)] \wedge \exists v \in \text{EPI}(Sp, w). [\neg \phi(v)].$

The key intuition behind (26) is that one can be fully certain that something is true based on the information available to them yet be aware that this evidence doesn't cut the threshold required for knowledge. In my sample, Japanese *sooda* belongs to hearsay evidentials that give rise to an inference of this sort. The semantics in (26) predicts that strong disavowals asserting $\neg \phi$ or knowledge that $\neg \phi$ will be banned, while weaker disavowals asserting that the speaker believes $\neg \phi$ will be allowed.

- (27) a. Japanese
 #Maria-wa Bahama-ni shunookeringu-ni it-ta **sooda** [=12a]
 Maria-TOP Bahaamas-to snorkeling-to go-PST **REP**
 ga it-tei-nai.
 but go-RES-NEG
 Intended: 'Maria went snorkeling on the Bahamas (I heard) but she didn't go.'
- b. $\llbracket (27a) \rrbracket^{c,g,w}$
 $= \exists x. \forall w' \in \text{CONTENT}(x, w). [\text{SNORKELING}(\text{Maria}, w')]$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w))$
 $\wedge \exists v \in \text{EPI}(Sp, w). [\text{SNORKELING}(\text{Maria}, v)]$
 $\wedge \exists v' \in \text{EPI}(Sp, w). [\neg \text{SNORKELING}(\text{Maria}, v')]$
 $\wedge \forall w'' \in \text{EPI}(Sp, w). [\neg \text{SNORKELING}(\text{Maria}, w'')].$

The semantics correctly predicts that disavowals explicitly talking about knowledge, such as 'I know that $\neg \phi$ ', lead to a semantic contradiction. Disavowals simply asserting $\neg \phi$ will be infelicitous for pragmatic reasons, assuming something like the norm of assertion (cf. also Footnote 22): in order to make a claim that $\neg \phi$, you have to know it, which is in conflict with the unsettledness of knowledge conveyed by the evidential. Weaker disavowals, including 'I believe $\neg \phi$ ', are felicitous, because belief does not entail knowledge and there is no semantic conflict. I also assume that knowledge does not entail belief, and that the unsettledness of knowledge does not entail unsettledness of belief (that there is at least one ϕ -world and at least one $\neg \phi$ -world compatible with the speaker's doxastic alternatives). Semantic literature often views doxastic predicates like *believe* and factive predicates like *know* as scalar alternatives, in which case 'knows' always entails 'believes'. However, recent research argues that factive predicates have a more involved semantics (Dj  rv 2023; Kratzer 2002), in line with philosophical research on the concepts of 'knowledge' and 'belief' (Dudley and Vogel 2023; Nagel 2017; Phillips et al. 2021). I assume that some hearsay evidentials are sensitive precisely to this distinction.²¹ Note that the semantics above also correctly predicts that follow-ups asserting the truth of ϕ will be infelicitous, as they are in direct conflict with the requirement that the speaker does not know whether the prejacent is true.

21. An alternative analysis could simply argue that that *sooda* encodes high credence without full certainty, but I prefer to maintain a categorical distinction rooted in justification requirements of different attitudes.

- (28) a. Japanese
 #ame-ga fut-teir-u-**sooda**-shi, jissaini fut-teir-u.
 rain-NOM fall-PROG-NPST-**REP**-CONJ really-fall-PROG-NPST
 Intended: ‘Is is raining (I heard), and it really is raining.’
 (adapted from Matsubara 2017:67, ex.23)

b. $\llbracket (28a) \rrbracket^{c,g,w}$
 $= \exists x. \forall w' \in \text{CONTENT}(x, w). [\text{SNORKELING}(\text{Maria}, w')]$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w))$
 $\wedge \exists v \in \text{EPI}(Sp, w). [\text{SNORKELING}(\text{Maria}, v)]$
 $\wedge \exists v' \in \text{EPI}(Sp, w). [\neg \text{SNORKELING}(\text{Maria}, v')]$
 $\wedge \forall w'' \in \text{EPI}(Sp, w). [\text{SNORKELING}(\text{Maria}, w'')].$

Previous analyses of Japanese *sooda* don’t capture the relevant data. The analysis in (McCreedy and Ogata 2007) is too weak: it treats *sooda* as an ordinary speech report without epistemic restrictions and incorrectly predicts that *sooda* will behave like *sollen*, allowing complete non-endorsement as well as endorsement. The analysis in Matsubara (2017), on the other hand, is too strong. In line with Kratzer (2012), it treats *sooda* as an informational modal that must be compatible with the speaker’s epistemic state, thus collapsing the distinction between knowledge and belief, and incorrectly predicts that all disavowals will be banned, not only those that assert $\neg\phi$ but also those that talk about the speaker’s belief.

Presence of belief Finally, some hearsay evidentials have been argued to ban all disavowals and convey that the speaker endorses the prejacent. I don’t have access to languages with such expressions, but, as discussed in Section 2.2, their existence is not surprising if we consider the behavior of hearsay evidentials against a larger context of speech reports. English has clause-taking predicates conveying that the speaker endorses the content of a reported claim, such as *be right*, and I propose that hearsay evidentials that import full endorsement can be modelled along similar lines. Absent relevant data, I can only speculate about the availability of follow-ups asserting the truth/falsity of ϕ .

- (29) a. Let $\text{CONTENT}(x, w)$ be the propositional content associated with the source x in w , R an acquaintance relation and $\text{DOX}(Sp, w)$ the set of worlds compatible with the speaker’s beliefs at w .
- b. $\llbracket \text{HEARSAY}_{\text{BELIEF}} \phi \rrbracket^{c,g,w} = \exists x. \forall w' \in \text{CONTENT}(x, w). \llbracket \phi \rrbracket^{c,g,w'}$
 $\wedge \exists R. R(Sp, w, \text{CONTENT}(x, w))$
 $\wedge \forall v' \in \text{DOX}(Sp, w). \llbracket \phi \rrbracket^{c,g,v'}$.

The literature treats such hearsay evidentials as modals making reference to realistic conversational backgrounds (Kratzer 2012; Matthewson et al. 2007). My proposal is different in two ways. First, I treat those evidentials as making reference to beliefs rather than facts. Second, the analysis retains a hearsay component in the semantics. It thus allows for a possibility of having evidentials that specifically require the existence of a report (rather than indirect evidence more generally) that the speaker believes, in parallel to veridical communicative predicates such as *be right* or *confirm*. Future research will show which account is on the right track.

3.4 Recap

Let's take stock. By using familiar diagnostics of intensional environments, this section establishes that at least some hearsay evidentials introduce intensional domains. Unlike bare sentences without evidentials and like speech reports with communicative predicates, these evidentials ameliorate contradictions and block presuppositions, which is unexplained under any extensional account. I offer an account of such hearsay evidentials as operators quantifying over contents of reports and propose that some of them instantiate hybrid attitudes, which in addition to encoding the existence of a speech report about the prejacent conventionally encode the speaker's attitude towards its truth. As shown already in Section 2.2, there is more variation in the shade of the speaker's attitude than previously assumed. It is not just the difference between endorsement vs. non-endorsement, but a more subtle distinction between the lack of attitude (German *sollen*, Turkish *mlş*), the lack of knowledge (Japanese *sooda*), unsettledness (English *apparently*, Dutch *schijnen*) and the presence of belief. Importantly, what I propose is the categorical difference in the type of attitude evidentials can track, but not gradient differences in credences they give rise to. Hearsay evidentials, like speech reports more generally, can be associated with different levels of commitment based, for example, on the source's reliability, but such pragmatic calculations needn't be part of the semantics proper (cf. McCready 2015).

Krawczyk (2012:294) argues that hearsay evidentials are never fully neutral with respect to endorsement because even those hearsay evidentials that allow strong disavowals don't allow two consecutive evidentials with contradictory prejacentes. (30) illustrates:

(30) German

#Es **soil** am Ferenginar geregnet haben. Es **soil**
it **REP.3SG.PRES** at.DEF Ferenginar rain.PTCP have.INF it **REP.3SG.PRES**
am Ferenginar nicht geregnet haben.
at.DEF Ferenginar NEG rain.PTCP have.INF
Intended: 'I heard that it rained on Ferenginar and that it didn't.'

As discussed by Faller (2002:247) and Murray (2010:28-29), two consecutive hearsay evidentials are by default interpreted as having the same source due to discourse subordination, and utterances like (30) attribute a contradiction to another party. I propose that such utterances are also infelicitous because of the discourse dynamics: in the absence of alternative QUDs, only the prejacent of evidential utterances is at-issue and (30) is interpreted as the speaker endorsing two contradictory prejacentes. I come back to this issue in Section 5 and for now conclude that there are endorsement-neutral evidentials, after all.

My overarching goal was to model the semantics of hearsay evidentials in a way that captures some aspects of the non-endorsement puzzle, specifically, the principled difference between hearsay vs. non-hearsay and the variation within hearsay. However, there is an aspect of the behavior of evidentials that an analysis of this sort doesn't capture: their discourse function. In the next section I turn to updates with hearsay evidentials, however, as I argue in detail in Section 5, none of the present accounts, whether modal or non-modal, explains the discourse dynamics associated with overt non-endorsement.

4 Speech acts with hearsay evidentials

The non-endorsement puzzle has featured prominently in the debates on the proper treatment of evidentials. In particular, the existing literature often views the differences between evidentials that do not license disavowals vs. those that do as a reflection of the putative categorical difference between modal vs. non-modal evidentials. I believe that this framing has obscured the similarities between evidentials and communicative predicates, and also unnecessarily conflates the modelling of the core semantics of evidentials and the modelling of the discourse effects of utterances that contain them. In this section, I review two accounts of non-endorsement wherein speech acts with hearsay evidentials in some ways deviate from standard assertions: by [Murray \(2010, 2014, 2017\)](#) and by [Faller \(2019\)](#). They both advocate the illocutionary view on evidential meaning such that evidentials are extensional operators directly interacting with the structure of discourse, however, each account is in principle compatible with intensional semantics and my focus below is on the discourse dynamics.

4.1 Hedges in update semantics

[Murray \(2010, 2014, 2017\)](#) argues that evidentials in languages like Cheyenne (Algonquian: Montana, USA) behave in the same way as parenthetical constructions (e.g., appositives and non-restrictive relative clauses) in languages like English: they convey novel backgrounded information that isn't part of the main point of an utterance. Evidentials are argued to be especially similar to the so-called slifting parentheticals (31; term due to [Ross 1973](#)), which have been first argued to have an evidential function by [Rooryck \(2001a,b\)](#) (see also [Simons 2007](#)).

(31) There is a new wormhole to the Gamma Quadrant, **I heard / I saw / I take it**.

Murray advocates an architecture of discourse wherein the difference between at-issue vs. not-at-issue information is associated with updates of different types, couched in Update with Centering ([Bittner 2011](#)). At-issue content, including the prejacent of evidential utterances and the main clause in slifting constructions, constitutes a proposal to update the common ground (cf. also [Anderbois et al. 2015](#); [Farkas and Bruce 2010](#); [Potts 2005](#)). It is the focus of discourse action: subject to conversational negotiation, it may or may not be accepted. Regular assertions consist of several steps. One of them is the act of presenting the at-issue proposition ϕ and creating a corresponding discourse referent to make it available for anaphora. Another is the proposal to update, formalized through a special illocutionary relation that induces an ordering on the context set that is defined by the clause type and may also be lexically specified by the construction. Only successful assertions result in a new, updated common ground. Not-at-issue content, including the evidential signal (information about how the speaker acquired the prejacent), is associated with direct, not-easily-negotiable updates. Regardless of how the conversation proceeds with respect to the at-issue contribution, the new common ground will be restricted to worlds where not-at-issue information holds. Thus, with a vanilla assertion *There is a new wormhole to the Gamma Quadrant* the speaker presents the prejacent proposition and proposes to add it to the common ground, given that unmarked assertions are expressed via declarative sentences whose illocutionary relation puts ϕ -worlds on top. With a slift like (31), however, the speaker also signals how they came by this information (specified by the parenthetical clause).

Assertions are canonically associated with the speaker's endorsement.²² Various expres-

22. Various authors have argued that assertions require truth, knowledge, justified belief, or commitment, stem-

sions, including epistemic modals, evidentials and slifting parentheticals, allow one to weaken that endorsement, which is usually referred to as ‘hedging’ (Benton and van Elswyk 2020). Thus, the speaker uttering (31) is taken to have some credence in the prejacent, but not necessarily as being fully certain (see references and discussion in Koev 2022). In Murray’s system, the level of speaker’s commitment is linked to the construction-specific illocutionary relation. The hedging effect of slifting parentheticals arises because the proposed ordering ranks $\Diamond\phi$ -worlds higher, and not just ϕ -worlds as with assertions. Thus, by uttering (31) the speaker presents the prejacent proposition, creates a direct update that restricts the incoming context set to the content of the parenthetical clause and signals a commitment to $\Diamond\phi$.

Hearsay evidentials present a problem: as discussed extensively throughout this paper, at least some such evidentials allow the speaker to fully void endorsement and express a belief $\neg\phi$. If assertions require endorsement, it follows that sentences with disavowals do not assert the prejacent. Murray argues that utterances with hearsay evidentials differ from other expressions in the nature of the illocutionary relation they introduce. Specifically, the hearsay relation does not rank ϕ - or $\Diamond\phi$ -worlds higher, but simply restricts the proposed update to the current context set. The key idea is that all utterances—with evidentials or without—have the same underlying structure, but distinct operators may hard-wire different discourse effects and thus lead to different commitment levels by virtue of the illocutionary relation they introduce.

One of the disadvantages of this system, and most dynamic frameworks, is that conversational updates are directly linked to the semantics of expressions that give rise to them. We have to stipulate that it is the hearsay evidential that leads to updates compatible with disavowals while non-hearsay evidentials will always induce a degree of credence in the prejacent and therefore are incompatible with disavowals. In other words, we have no principled explanation of the differences between hearsay vs. non-hearsay in terms of non-endorsement, one of the desiderata for a theory of non-endorsement outlined in Section 2.

There are also other, related, concerns. Commitment in this system is treated as an arbitrary property of specific constructions: appositives induce commitment to the prejacent, slifting parentheticals to the modalized prejacent, and hearsay evidentials to nothing at all. Putting evidentials aside for a moment, there are systematic differences within constructions that give rise to not-at-issue entailments. Thus, Pottsonian supplements do not influence the speaker’s credence. (32) illustrates for an *as*-parenthetical in English: compared to standard clausal embedding (3), these constructions do not allow disavowals.²³

(32) NO NON-ENDORSEMENT WITH AS-PARENTHETICALS

#As Sisko claimed, there is a new wormhole to the Gamma Quadrant, but I don’t believe it/but I know it’s false/but I’m skeptical.

Slifting parentheticals, on the other hand, present a more interesting case. Koev (2022) shows that commitment isn’t tied to the construction itself and varies based on the content of the slifted clause, including the embedding verb, but assumes that some degree of credence is always present. This remains an underexplored empirical question—for example, many people find (33a) acceptable, especially in comparison to the utterly infelicitous (32). However, disavowals are never acceptable with first-person mental attitude ascriptions (33b), so we observe

ming from sincerity conditions, constitutive norms, conversational maxims or special speech-act operators (see extensive discussion and references in Goldberg 2020; Marsili 2024; Pagin and Marsili 2025). The exact source and nature of endorsement is unimportant here as long as we expect that under normal circumstances speakers are taken to believe the content of their unqualified (bare, unhedged) assertions and that they can be held accountable for that content in later discourse.

23. There is still little cross-linguistic research on parenthetical constructions, so I limit my discussion to English.

the general contrast between hearsay vs. non-hearsay in the availability of non-endorsement discussed in Section 2.1 (with the same caveat about tense as brought up in Footnote 9).

(33) SLIFTING PARENTHETICALS AND NON-ENDORSEMENT

- a. ??There is a new wormhole to the Gamma Quadrant, **Sisko claimed**, but I don't believe it.
- b. #There is a new wormhole to the Gamma Quadrant, **I think / I gather / I was led to believe**, but I don't believe it.

Murray's system treats different constructions that contribute not-at-issue meaning on a par and therefore does not predict systematic differences found within them (though different properties can be stipulated on a construction-by-construction basis): hearsay vs. non-hearsay, *as*-parentheticals vs. slifting parentheticals. Furthermore, it does not take into consideration possible lexical variation within speech reports with respect to endorsement, whether with hearsay evidentials or communicative predicates (though it is possible to express these notions using the illocutionary relation). Koev (2022) advocates a more refined view on parenthetical constructions in English such that some differences between them stem from their syntactic height with respect to an illocutionary force operator in the root clause. However, he does not discuss evidentials in detail and his system explicitly excludes the possibility of disavowals such as (33a) (Koev 2022:139), so I will not discuss its applicability to evidentials.

4.2 Presentational speech acts

Another prominent account of non-endorsement is due to Faller (2019), developed for Cuzco Quechua (Quechuan: Peru).²⁴ Faller assumes an evidential model of discourse within the general framework of the Table model (Farkas and Bruce 2010). The key idea is as follows. The conversational scoreboard consists of several components, including the common ground, the Table for recording issues under discussion and the sets of discourse commitments for each interlocutor. Commitments may come in different flavors depending on how the interlocutors acquired them (cf. Gunlogson 2003; Malamud and Stephenson 2015), and different evidentials can be associated with commitments of different type; see also Bhadra (2020); Faller (2023, 2024); Martínez Vera (2024); Meriçli (2016); Northrup (2014) for similar approaches to evidentials.

Standard assertions in this framework are taken to put ϕ on the Table and add it to the speaker's discourse commitments. If this conversational move is accepted, the addressee's discourse commitments in the next stages of discourse will contain ϕ as well, the Table will be cleared and ϕ will become part of the common ground. Not-at-issue information is not directly represented. According to Faller, non-hearsay evidentials add the prejacent to the speaker's Truth Commitments (propositions whose truth the speaker is committed to) and additionally specify that the speaker has adequate evidence for it by updating the respective commitment set.

Hearsay evidentials, on the other hand, are argued to be not assertions, but speech acts of presentation. The reasoning behind it is very similar to the one discussed in Section 4.1: at least some hearsay evidentials, including Cuzco Quechua *=si*, require no endorsement and allow strong disavowals. Evidential contribution is taken to be always not-at-issue and as such

24. As Faller (2019) herself points out, the account in (Faller 2002) is an earlier version of the same general idea couched in a different framework, so I will only discuss the more recent version.

off the Table, which means that utterances with such evidentials cannot be assertions, since assertions require some endorsement of their at-issue content. Capitalizing on Goffman’s (1979) distinction between different discourse roles a speaker may perform, Faller proposes the following solution (cf. also Pancheva and Rudin 2019). In regular utterances, the speaker is often both the *animator* (the person who physically produces the sentence) and the *principal* (the person whose point of view is conveyed). However, hearsay evidentials allow us to report positions that we don’t share and can disagree with in subsequent discourse. Faller argues that this is at the heart of non-endorsement puzzle: hearsay evidentials put the prejacent on the Table, add it to the propositions for which the speaker has hearsay evidence but don’t signal anything about the speaker’s attitude. Instead, they only indicate that the principal, a person distinct from the current speaker, has publicly committed to it at a previous point. The current speaker will be taken to endorse the prejacent by default due to the Collaborative Principle, which dictates that any disagreement with expressed content must be stated right away (Walker 1996). However, because endorsement is not hard-wired, the speaker can signal their disagreement via disavowals without sounding contradictory (unlike with assertions).

The system sketched above provides a straightforward explanation of the difference between hearsay vs. non-hearsay, since only hearsay evidentials can introduce the point of view of others. However, it is not unproblematic when we try to extend it to the non-endorsement puzzle more generally (which isn’t Faller’s explicit aim). My main concerns are as follows.

First, it is unclear what natural language phenomena could be analyzed as involving the speech act of presentation. As discussed in Section 2, many types of speech reports allow non-endorsement. Some of them, like slifting parentheticals (33a), are usually taken to contribute not-at-issue information and thus could be assimilated to hearsay evidentials of the Cuzco Quechua type. However, as discussed in previous section, different constructions seem to allow different levels of endorsement—where does that come from? And why *as*-parentheticals, even when they contain a speech report, ban disavowals altogether (cf. 32)? Does that mean that distinct lexical items give rise to different speech acts in an arbitrary way? I am not arguing with the idea that such speech acts might exist, but it would be interesting to explore possible constraints on the mechanism. It’s also unclear which declaratives can be interpreted as presentations. On the one hand, *I heard that there is life on distant planets, Although I disagree, let’s discuss* sounds felicitous. On the other, *There is life on distant planets. Although I disagree, let’s discuss* sounds infelicitous. Both are declaratives, but the first one contains a speech predicate whose function is very similar to hearsay evidentials. However, evidentials are argued to be fundamentally different from speech predicates, so it’s unclear how to get out of this conundrum.

Second, the system can only account for hearsay evidentials that don’t encode anything about the speaker’s attitude—it predicts that the speaker can go as far as asserting $\neg\phi$, which is the case in Cuzco Quechua. However, as we have seen in Section 2.2, hearsay evidentials vary in the type of information they encode. One solution would be to argue that all such evidentials are inherently modal, which will be circular given that some evidentials that allow strong disavowals, like German *sollen*, are modal and it would be good to have a system that accounts for discourse updates with such evidentials as well. Another solution would be to associate graded commitments directly with distinct constructions and/or lexical items (cf. Northrup 2014; Scheffler and Malamud 2023), but then it’s unclear what the source of commitment is (cf. also Rudin 2022).

Finally, Faller’s account predicts that somewhere in the chain of speech reports there is always a party that endorses the prejacent of a hearsay claim, even if it’s not the current speaker (the same goes for AnderBois 2014; the problem is also replicated in Smirnova’s (2012) belief-

based account). This leads to the puzzle of recursive non-endorsement.

At least some hearsay evidentials can be used to report claims explicitly disavowed by people making them, for example, if their own evidence is based on hearsay. (34) illustrates for German and Turkish (which semantically don't encode the speaker's attitude towards the prejacent; Section 3.3).

(34) **Recursive non-endorsement**

Lisa says that, based on a notification she just received, her paper was accepted to a prestigious conference. She doesn't believe it, she says her paper was so bad. You yourself have no opinion, but discuss it with other colleagues nonetheless, since none of you have gotten any notifications yet and gossip gives you something to do.

a. German

Sein Konferenzbeitrag **soll** angenommen worden
 his/her conference.abstract.NOM.SG **REP.3SG.PRES** accept.PTCP become.PTCP
 sein.
 be.INF
 'Her paper is supposed to have been accepted.'

b. Turkish

makale-si kabul edil-**miş**
 paper-3 acceptance make-**IND**
 'Her paper was accepted (**I heard**).'

One could argue that in (34) endorsement lies with the people who send out conference notifications. First of all, we may run into the problem of attributing commitments to inanimate entities if such notifications are issued automatically (cf. [Anand and Hacquard 2014](#); [Anand et al. 2017](#) on inanimate subjects of communicative verbs). More generally, however, Faller's account requires that hearsay evidentials only report sincere assertions, or at least utterances taken as such by the current speaker. This incorrectly predicts that hearsay evidentials can't report cases of known lies (they can), and imposes too many constraints on the denotation of evidentials. I argue for a division of labor instead. It is clear that considerations about the original source's general reliability as well as certainty influence the type of update the speaker intends to make with a hearsay utterance (cf. [McCready 2015](#)). However, it is also possible to have a semantics that remains agnostic about beliefs and/or commitments made by third parties. I have sketched an intensional account in Section 3, but this is not the only option and Murray's proposal captures this aspect of hearsay evidentials as well, since it doesn't attribute commitments to anyone other than the speaker.

4.3 Recap

This section discussed two commitment-based approaches to the non-endorsement puzzle. The account in ([Murray 2010, 2014, 2017](#)) treats evidentials as hedges in update semantics but doesn't predict the systematic differences between hearsay vs. non-hearsay evidentials or between different parenthetical constructions. The account in ([Faller 2019](#)) treats utterances with hearsay evidentials as speech acts of presentation, but doesn't specify how widely applicable the mechanism is and faces the problem of recursive non-endorsement.

Despite their architectural differences, both accounts rely on the following key assumption: only the prejacent can be at-issue in utterances with evidentials. It is this assumption that leads to the idea that utterances with hearsay evidentials cannot be regular assertions (cf. also general

discussion in [Bary and Maier 2021](#)). However, neither these accounts nor others on the market actually explain how discourses with disavowals are used or which QUDs they address. I turn to this pivotal question in the next section.

5 Discourse dynamics

The existing literature focuses on the modeling question: how to best formulate the *denotation* of hearsay evidentials to make non-endorsement non-contradictory. As discussed in previous sections, there are various solutions available. At the same time, there is an overlooked dimension to the puzzle: the overall *discourse function* of hearsay utterances with explicit non-endorsement. [Faller \(2019\)](#) meticulously shows that utterances with Cuzco Quechua *=si* make the prejacent at-issue, for example, they participate in veridical discourse relations and can be used to address or resolve the QUD.²⁵ However, she focuses on hearsay simpliciter. The ubiquitous underlying assumption is that the discourse function of evidential utterances is always the same, with or without disavowals, and that the evidential signal is conventionally marked as not-at-issue, backgrounded information. While modal approaches to evidentials usually don't talk about their discourse functions at all, this assumption is especially important in those frameworks where evidentials are taken to directly interact with the structure of speech acts, giving rise to construction-specific discourse updates. The key idea, backed by independent assumptions about clause typing, is that hearsay evidentials always give rise to the same type of update. I propose a different take: hearsay evidentials behave like communicative predicates in that they have parenthetical uses (by default) and non-parenthetical uses (marked, signalled by disavowals). They appear in different discourses with different QUD structures and needn't be modelled in the same way.

Building on [Korotkova \(2020\)](#) and using [Hunter's \(2016\)](#)'s analysis of discourse parentheticals as a guiding parallel, I argue that the evidential signal does not have a fixed at-issue status and that disavowals appear precisely in those discourses that make more than the prejacent at-issue. For those hearsay evidentials that don't hard-wire belief, endorsement is then derived as a Relevance Implicature (cf. [Krawczyk 2012](#); [Hunter 2016](#)). While taking the discourse dynamics of evidential utterances seriously, this view avoids the problems faced by commitment-based accounts. Another advantage is that it only makes an independently motivated assumption that evidential not-at-issueness is best understood in terms of QUD-relevance (cf. also [Koev 2022](#) on parenthetical constructions). It is therefore orthogonal to the modal vs. non-modal dichotomy and is compatible with a variety of approaches to the semantics of evidentials, including, but not limited to, the analysis sketched in Section 3 or commitment-based accounts discussed in Section 4.

5.1 Evidential (not-)at-issueness

Not-at-issue content, including classical presuppositions and various parenthetical constructions, is informally understood as backgrounded, peripheral information that does not bear on the main point of an utterance. Depending on the specific theoretical assumptions about the architecture of discourse, this intuition can be cashed out in different ways, each associated

25. [Faller \(2019:18-19\)](#) further suggests that the QUD-relevance of the prejacent is an argument against Murray's analysis of hearsay evidentials, where the prejacent isn't intended to be added to the common ground. This is an issue of specific implementation rather than general architecture: while Murray's framework doesn't make explicit reference to the QUD structure, it may be adjusted accordingly without sacrificing its core assumptions about discourse updates (see also discussion in [Snider 2017](#)).

with distinct empirical diagnostics (see overview in [Koev 2018](#)). The literature on evidentiality unanimously treats the evidential signal as a not-at-issue contribution, and in approaches discussed in Section 4 at-issueness is understood via potential to update the common ground. Much as with parenthetical meaning across the board, not all diagnostics lead to conclusive results when it comes to diagnosing evidential not-at-issueness. For example, the widely used assent/dissent test ([Papafragou 2006](#)) does not distinguish between anaphoric potential (possibility to be targeted by responses like ‘That’s surprising’) and disagreement (possibility to be targeted by responses like ‘That’s not true’ or ‘You’re mistaken’), while evidentials pattern like other self-attributions and are not compatible with denials regardless of their linguistic shape ([Korotkova 2016a](#)). As discussed in detail by [Korotkova \(2020\)](#), evidentials are best modeled using the QUD notion of at-issueness (cf. also [Koev 2022](#) on parenthetical constructions). QUD at-issueness defines the main point point of an utterance via its relevance to the current QUD, and is diagnosed via question-answer pairs ([Simons et al. 2010, 2017](#); [Tonhauser et al. 2013](#)). In this section, I refine the question-answer test for evidentials by taking into account the previously unnoticed PREJACENT UNSETTLEDNESS constraint.

Evidential utterances are most naturally interpreted in a way that the evidential signal is not addressing the QUD. (35) illustrates for a range of hearsay evidentials, including English *apparently*. In the discourses below, it is clear that the speaker is upset about the prejacent proposition (which they take to be true or at least possible). The fact that they acquired it via hearsay is additional, by-the-way information that does not bear on the question.

(35) **QUD about prejacent**

You learn via the grapevine that your friends are relocating.

A: Why are you upset?

B: **Apparently**, my friends are moving away. I will miss them.

a. German

Meine Freunde **sollen** laut Sarah wegziehen. Ich werde sie
my friend.PL **REP.3PL.PRES** according.to Sarah move.INF I will they
vermissen.
miss.INF
‘According to Sarah, my friends are moving away. I will miss them.’

b. Turkish

Arkadaş-lar-ım taşın-ıyor-**muş**. Onları özley-eceğ-im.
friend-PL-1 move-PRES-**IND** they miss-FUT-1
‘My friends are moving away (**I heard**). I’ll miss them.’

c. Japanese

Tomodachi-ga moosugu hikko-su **sooda**. Karera-ga i-nai-to samishii.
friend-NOM soon move-PRES **REP** they-NOM be-NEG-COND lonely
‘(My) friends are moving soon (**I heard**). I’ll feel lonely without them.’

d. Dutch

Mijn vrienden **schijnen** verhuisd te zijn. Ik ga ze missen.
My friend.PL **REP.3PL.PRES** move.PTCP to be.INF I FUT they miss.INF
‘My friends are moving (**I heard**). I’ll miss them.’

The literature further assumes that evidentials are *conventional* triggers of not-at-issue content, just like the COMMA operator in *as*-parentheticals and appositives has been argued to signal that the content within its scope must be not-at-issue ([Potts 2005](#)). Building on [Ko-](#)

rotkova (2020) and Koev (2022), I propose a different, more flexible view: the evidential signal is backgrounded by default, but can become foregrounded in discourses that allow it. Additionally, some—but not all—evidentials mark their prejacent as at-issue.

Unsettledness Discourses like (36) are often used to argue that evidentials, unlike communicative predicates, cannot answer questions about evidence and therefore must be not-at-issue.²⁶

(36) **QUD about evidence (take 1)**

Question: What makes you think it will rain?

Dutch

#Het schijnt te regenen.

this REP.3SG.PRES INF rain

Intended: ‘It’s said that it will rain.’

(Korotkova 2020:16)

Such examples merit further scrutiny. Even with communicative predicates, answers that simply repeat the content of the question are uninformative: for example, *I’ve been told that it would rain* is infelicitous in (36). The acceptability improves when the prejacent is marked (i) as presupposed with an anaphor *I’ve been told so*, or (ii) as given with focus marking on the verb *I’ve been TOLD that it would* (when available, focus also improves the acceptability of evidential utterances; Korotkova 2020:16). Based on data similar to (36), Bary and Maier (2021) argue that *according to* adverbials are at-issue, however, replies with *according to* show the same pattern as verbal speech reports: they are odd without special means such as focus. In Section 3.3, I model the semantics of hearsay evidentials along the same lines as *according to*, and I maintain that these two types of constructions have more in common than previously thought, including in terms of at-issueness.

I propose that discourses in (36) are infelicitous not because of at-issueness per se, as it has been claimed in the literature, but because at least some hearsay evidentials require that their prejacent isn’t already settled, just like English *claim* or *posit*. Consider the contrast between (37) and (38) below.

(37) **Prejacent not settled**

There have been various attempts to develop a vaccine against Lyme disease. While many of them have proved unsuccessful, there is always hope. After reading an article documenting new ongoing trials, you say:

a. The article ✓**states**/✓**reports**/✓**claims**/✓**posits** that the new Lyme vaccine is effective.

b. ✓**Apparently**, the new vaccine is effective.

c. German

Das neue Impfstoff **sell** gegen Borreliose wirksam sein.

DEF new vaccine REP.3SG.PRES against Lyme effective be.INF

‘The new vaccine is effective against Lyme (I heard).’

26. Unlike their projective behavior or anaphoric potential, the answerhood diagnostic has not been used as widely with evidentials; I’m aware of the data like (36) for Aymara (Martínez Vera 2024), Cuzco Quechua (Faller 2019), Georgian (Korotkova 2020), Gitksan (Bary and Maier 2021), Japanese [redacted for anonymity], Kiche [redacted for anonymity] and Korean (Lee 2013).

- d. Turkish
Yeni aşı çok etkili-y-miş
new vaccine very effective-COP-IND
'The new vaccine is very effective (I hear).'
- e. Dutch
Het nieuwe vaccin schijnt effectief te zijn!
DEF new vaccine REP.3SG.PRES effective inf be.INF
- f. Japanese
wakuchin-wa kookateki-da sooda
vaccine-TOP effective-COP sooda
'The vaccine is effective (I hear).'

(38) **Prejacent settled**

Vaccine hesitancy is on the rise and, to counteract that, media publish various facts about vaccines' efficiency. That said, we both already agree that vaccines are one of the best inventions of humankind and are on top of relevant science being geeks and all, so there is nothing new for us in those reports. You pick up a newspaper and say:

- a. The article ✓states/✓reports/#claims/#posits that flu vaccines are effective.
- b. #Apparently, flu vaccines are effective.
- c. German
#Grippeimpfungen sollen wirksam sein.
flu.vaccine.PL REP.3PL.PRES effective be.INF
'Flu vaccinations are effective (I heard).'
- d. Turkish
#Aşı-lar çok etkili-y-miş
vaccine-PL very effective-COP-IND
'Vaccines are very effective (I hear).'
- e. Dutch
#De grieprik schijnt effectief te zijn.
DEF flu.shot REP.3SG.PRES effective to be.INF
'The flu vaccine is effective (I hear).'
- f. Japanese
#wakuchin-wa kookateki-da sooda
vaccine-TOP effective-COP sooda
'The vaccine is effective (I hear).'

The key difference between (37) and (38) is whether the issue raised by the prejacent is settled.²⁷ The prejacent of (38) will be considered not-at-issue in those frameworks where at-issueness is equated with update potential (Anderbois et al. 2015; Farkas and Bruce 2010; Murray 2014): only at-issue content can update the common ground, and all content that is not intended as a proposal to update is automatically not-at-issue. Importantly, according to the QUD notion of at-issueness, there is no requirement that content not intended as a proposal to update is not-at-issue. This difference is important in disentangling which diagnostics work

27. Note that the contrast is not about non-endorsement: all evidentials in (37) and (38) are compatible with, and are often interpreted as, the speaker's endorsing the prejacent.

best for evidentials.

Korotkova and Anand (2025) argue that *claim* is infelicitous in discourses like (38) because its function is more than reporting a previous communicative act—it specifically refers to a discourse stage when a proposal has been made but has not been accepted (yet or at all) by the relevant community. Many other speech verbs, including the neutral *state* and *report*, place no such requirements and are acceptable in the right context even if the complement clause does not contribute new information (38a). I propose that similar factors are at play with hearsay evidentials. They are all felicitous in (37), which shows their general compatibility with such contexts. However, as their infelicity in (38) illustrates, they require that the issue raised by the prejacent is not settled among the interlocutors (more formally, they can be taken to presuppose $[\phi \notin cg \wedge \neg\phi \notin cg]$, where *cg* is the common ground). Some evidentials can be used in (38) sarcastically ($\approx$ What else is new?), especially those that generally allow a mirative interpretation, such as Turkish *mİş* or English *apparently*. Even in those cases, there is a marked contrast with the neutral utterances in (37). Such data further underscore the need for a more detailed comparison between evidentials and speech predicates—we might be able to find evidentials that pattern like *report* rather than *claim*. Importantly, such data also allow us to rethink the answerhood test for evidentials.

At-issueness I propose to use the following question-answer pairs to test whether the evidential signal may be at-issue on its own, not in conjunction with the prejacent. Importantly, in the examples below the prejacent is not already settled in previous discourse, so such examples allow us to disentangle that from QUD-relevance. In (39), the speaker is clearly endorsing the prejacent at least to some degree, but the reason for their being upset is not the party itself, but rather that they learned about it.²⁸

(39) **QUD about evidence (take 2)**

You learn via the grapevine that your friends are organizing a surprise birthday party—for you!

Question: Why are you so upset?

- a. ✓ I heard that my friends are throwing a surprise party for me. Such a bummer. I love surprises.
- b. #My friends are throwing a surprise party for me, I learned/heard. Such a bummer. I love surprises.
- c. #My friends, as I just heard, are throwing a surprise party for me. Such a bummer. I love surprises.
- d. #According to the rumors/gossip, my friends are throwing a surprise party for me. Such a bummer. I love surprises.
- e. #Apparently, my friends are throwing a surprise party for me. Such a bummer. I love surprises.

28. Snider (2017:256-258) warns that *why*-questions might be tricky in probing at-issueness, since they ask for explanations and don't have simple, exhaustive answers. Yet, content that is conventionally marked as not-at-issue, such as presuppositions of hard triggers, never answers these (or any other) questions, so I maintain that we can use this diagnostic with caution.

f. German

#Meine Freunde **sollen** eine Party für mich organisieren. Echt
 my friend.PL **REP.3PL.PRES** INDEF party for me organize.INF total
 schade, ich bin ein Fan von Überraschungen.
 sadness I be.1SG.PRES INDEF fan of surprise.PL.DAT
 Intended: ‘I heard that that my friends are throwing a surprise party for me. What a
 pity, I am a fan of surprises.’

g. Dutch

#Mijn vrienden **schijnen** een verrassing voor me te organiseren. Zo
 my friend.PL **REP.3PL.PRES** INDEF surprise for me to organize.INF so
 jammer. Ik hou van verrassingen!
 pity I enjoy of surprise.PL
 ‘My friends are organizing a party for me (I heard). What a pity. I love surprises!’

h. Turkish

✓Arkadaş-lar-ım ben-im için sürpriz bir doğum gün-ü partisi
 friend-PL-1 I-GEN for surprise one birth day-ACC party
 düzenl-iyor-lar-**miş**. Kötü ol-du, sürpriz-ler-i pek sev-er-im.
 hold-PRES-PL-**IND** evil be-PST surprise-PL-ACC very like-PRES-1
 ‘I heard my friends are throwing me a birthday party. Such a pity, I love surprises.’

i. Japanese

✓Tomodachi-ga paatii-o hirai-tekure-ru **sooda**. Zannen-da. Watashi-wa
 friend-NOM party-ACC hold-BEN-PRES **REP** bummer-COP I-TOP
 sapuraizu-ga suki-da.
 surprise-NOM like-COP
 ≈ ‘I heard (my) friends are throwing me a party. A bummer. I like surprises.’

We can see that different constructions fall into two groups. I will start with English parentheticals. Koev (2022) proposes that slifting parentheticals and *as*-parentheticals cannot answer questions not because they are conventionally marked as QUD-irrelevant/not-at-issue (as previous literature assumes; Potts 2005; Murray 2014), but because it’s the main clause that has to always be QUD-relevant. Thus, if we take the parenthetical out in (39b,c), the discourses will still be odd, in line with Koev’s proposal. Both discourses can be improved if the information about the report—and thus the cause for bad mood—is added via a separate clause, something like *I saw it too early*. Several evidentials behave in the same way: English *apparently* (39e), German *sollen* (39f) and Dutch *schijnen* (39g). Importantly, English *according to* is also infelicitous when only the speech report is at-issue (39d), which is at odds with Bary and Maier’s (2021) analysis of it as an at-issue operator that only differs from communicative predicates in event structure—it should then behave as (39a). I propose that, just like English parentheticals in Koev’s analysis, these hearsay evidentials and English *according to* require their prejacent to be QUD-relevant, which in turn makes them infelicitous in (39).²⁹

On the other hand, non-parenthetical uses of communicative predicates (39a), Turkish *miş*

29. Further support for the idea that the existence of a speech report is conversationally not-at-issue with *according to* comes from its infelicity in discourses like (30), noted by Krawczyk (2012:338): #*According to Jake, it’s raining on Ferenginar and it’s not raining on Ferenginar*. Krawczyk argues that such sentences are infelicitous because they attribute a contradictory claim to one source, however, the corresponding speech report with a communicative verb sounds better: *Jake said that it’s raining on Ferenginar and that it’s not raining on Ferenginar*. I propose that this asymmetry is due to the fact that *according to* marks the prejacent as at-issue, and we end up with the speaker committing to two contradictory claims, which is infelicitous, just like with hearsay evidentials.

(39h) and Japanese *sooda* (39i) can serve as the sole main point of an utterance in a rich enough context, which shows that their prejacent needn't be conventionally marked as QUD-relevant. Importantly, even for Turkish and Japanese this is not the default interpretation of sentences in (39). Without the follow-up explaining that the speaker loves surprises and are upset specifically because they learned about it, corresponding sentences are odd in (39), supporting my proposal that the evidential signal is QUD-irrelevant by default.

Let us summarize. Current literature treats the evidential signal as obligatory not-at-issue because the most natural use of evidential utterances is such that the prejacent is at-issue. However, the QUD notion of at-issueness, best applicable to evidentials for independent reasons (Korotkova 2020), doesn't require that a particular piece of information is *conventionally* marked as not-at-issue, only that it is *conversationally* not-at-issue in relation to a given QUD structure, using Koev's apt terminology. There seems to be a general tendency in natural language to express attribution of information and grounds for belief as discourse-backgrounded—we see it with various constructions that have evidential restrictions, but also with speech and attitude predicates, especially with doxastics. As Simons (2007) and Hunter (2016) show, speech reports with communicative predicates are often interpreted parenthetically even without parenthetical syntax. The same seems to be true for *according to* adverbials. We can therefore conclude that the evidential signal for evidentials is not-at-issue by default and that it may become at-issue in some discourses that warrant it.³⁰ Zooming in on question-answer pairs that have been used to diagnose not-at-issueness in existing literature, I have shown that hearsay evidentials require that their prejacent is not settled in the previous discourse. When this novel constraint is accounted for, we see that we are dealing with a genuine case of variation. Some evidentials allow for the evidential signal to be the sole main point of an utterance. Some other evidentials, as well as *according to* adverbials, require the prejacent to always be QUD-relevant (just like the main clause of parentheticals in Koev's (2022) analysis). This flexible view on at-issueness predicts that even for those evidentials whose prejacent must be QUD-relevant, the evidential signal may be at-issue in conjunction with the prejacent in discourses with more than one QUD. In the next section, I argue that this is precisely what happens in cases of overt non-endorsement.

5.2 Disavowals in discourse

Those frameworks that model the discourse dynamics of evidential utterances implicitly assume that hearsay evidentials always make the same contribution, which, in turn, led to theories wherein the speech acts they perform are not standard assertions across the board. Capitalizing on Hunter's (2016) analysis of speech reports in discourse and on Korotkova's (2020) proposal that evidentials aren't always not-at-issue, I argue that hearsay evidentials perform different functions in discourses without vs. with overt disavowals and in that respect resemble speech reports with communicative predicates.

No disavowals It has been often observed that in the absence of explicit disavowals hearsay evidentials are taken to indicate that the speaker endorses the prejacent at least to some extent

30. I've only been talking about hearsay evidentials, which allow the speaker to distance themselves from a report, but one wonders if we can find instances of non-hearsay evidentials when the evidential signal is at-issue. Simons (2007) notes that it's much harder to get a non-parenthetical reading with first-person attitude reports (*I think, I conclude*) and it's also more difficult to diagnose intensionality with them, as they don't separate the speaker and the attitude subject. Similar issues arise with non-hearsay evidentials, and I will not attempt to develop a general theory of evidential at-issueness here.

(I come back to this issue in the next section). This is easy to predict if we assume that the evidential signal is not-at-issue in those cases. Assertions, including with hedges, signal a degree of endorsement. If the speaker chooses a construction where by default only the prejacent is at-issue, then they must intend to update the common ground with it. The evidential in those cases merely signals that the source for the prejacent is hearsay. The key intuition, however, is the same: only the prejacent is QUD-relevant and serves to move the conversation forward in some way. Consider the discourse in (35), repeated below as (40).

(40) *You learn via the grapevine that your friends are relocating.*

A: Why are you upset?

[=35]

B: Apparently, my friends are moving away. I will miss them.

The specific at-issue contribution depends on whether we're dealing with intensional or extensional operators (see discussion in Section 3). For intensional hearsay evidentials, including English *apparently*, the at-issue contribution can be a modalized proposition, e.g., $\Box\phi$.³¹ In Koev's (2022) framework for parenthetical meaning, the discourse updates of such evidentials can be treated in the same way as those of slifting parentheticals, to which evidentials are often compared. In the standard Table model, the discourse update of such utterances will correspond to the speaker's committing to $\Box\phi$ and putting it on the Table, signaling their intention to update the common ground with it. The rest proceeds as usual: the proposal may be accepted or not, the grounds for the claim may be challenged or not. My main point is that discourses with endorsement (and without disavowals indicating non-endorsement) can be analyzed as regular assertions in one's framework of choice.

Overt disavowals Discourses with disavowals are more interesting: as argued extensively in the literature, they cannot be regular assertions if the prejacent is the only at-issue contribution—the speaker cannot intend to update the common ground with something they don't believe. The underlying assumption in the literature is that hearsay evidentials are fundamentally different from speech reports with communicative predicates, which are taken to easily allow the existence of report to be at-issue. I propose instead that disavowals with hearsay in fact can be treated as assertions much in the same way as ordinary speech reports with overt non-endorsement. To see this, it's instrumental to understand the discourse functions of speech reports with overt disavowals in general, not only with evidentials.

In her SDRT-analysis of 'discourse parentheticals' (constructions that give rise to parenthetical meaning without dedicated parenthetical syntax), Hunter (2016) shows that the binary at-issue vs. not-at-issue distinction, which was developed for dialogue and as such not well-suited for discerning the internal structure of larger texts (Hunter and Abrusán 2017), does not capture the range of rhetorical relations such constructions can participate in in narratives. She argues that the main clause of a discourse parenthetical often provides relevant information via the Attribution relation and sometimes can enter into rhetorical relations of its own. She

31. Epistemic modals have been argued to make both the prejacent and the modalized prejacent at-issue (von Stechow 2011; Fintel and Gillies 2011; MacFarlane 2011; Roberts 2023), presumably putting either ϕ or $\Box\phi/\Diamond\phi$ on the Table. Such data aren't always easy to replicate for evidentials (though see Matthewson et al. 2007), which has often been taken as an argument against modal accounts for at least some evidentials (Faller 2019; Murray 2014). However, there is an important confound here: epistemic modals seem to allow such readings when they aren't exclusively speaker-oriented, giving rise to the disagreement problem often solved by relativist machinery (see extensive discussion in Rudin 2025a). Evidentials—or at the very least hearsay evidentials—don't give rise to the disagreement problem in the first place, which may be orthogonal to whether they are intensional operators semantically. I sidestep this issue here, since it is not crucial to my discussion.

specifically discusses cases that signal the lack of endorsement, including overt disagreement:

(41) **Speaker disagreement**

John didn't come to my party. Jill said he was out of town, but she's wrong. He's just avoiding me. (Hunter 2016:20)

(42) **Contrasting viewpoints**

John didn't come to my party. Jill said that he was out of town, but Bill said that he's just avoiding me. (So I don't know.) (Hunter 2016:21)

Such cases present a problem for approaches wherein only one proposition, either the one expressed by the main clause or by the complement, is at-issue. Hunter argues that in (41), (42) and other similar discourses the main clause is not parenthetical. Its content is at-issue, connected to another position via the Contrast relation. Its function is to indicate the existence of information pertaining to the issue at hand (possible explanations of John's absence), even if the speaker ultimately disagrees with it (41) or ends up having no beliefs at all (42). The key assumption here is that information may or may not be at-issue based on the discourses it is used in, in line with the observations in (Simons 2007). Main clauses of discourse parentheticals are often not-at-issue, possibly because the attribution of information is less important in many discourses than the information itself. However, overt disavowals signal a change from this default and those are the cases when the main clause is not interpreted parenthetically. I argue that overt non-endorsement with hearsay evidentials should be treated along similar lines.

Consider (43) and (44).

(43) **Speaker Disagreement**

After the winter break, we're discussing who went where on vacation. We're now talking about Ana. You haven't spoken with her personally, but have heard various gossip.

- a. **Apparently** she went snorkeling in the Bahamas, though I don't believe it, she hates the heat and she said she has no money.

- b. German

Ehrlich gesagt habe ich keine Ahnung. Sie **soll**
honestly say.PTCP have.1SG.PRES I NEG.INDEF opinion she **REP.3SG.PRES**
auf den Bahamas geschnorchelt haben, aber das glaube ich
at def Bahamas snorkel.PTCP have.INF but this believe.1SG.PRES I
nicht. Sie hasst Hitze und hat mir auch gesagt, sie
NEG she hate.3SG.PRES heat and have.3SG.PRES me also say.PTCP she
hätte kein Geld.
have.3SG.SUBJ NEG.INDEF money
'Honestly, I've no idea. I heard she went snorkeling in the Bahamas, but I don't believe it. She hates the heat and also told me that she had no money.'

c. Dutch

Eerlijk gezegd weet ik het niet. Ze schijnt te hebben
 honestly say.PTCP know.1SG.PRES I this NEG she **REP.3SG.PRES** to have.INF
 gesnorkeld op de Bahama's, maar dat geloof ik niet. Ze
 snorkel.PTCP on DEF Bahamas but that believe.1SG.PRES I NEG zhe
 heeft een hekel aan warm weer en zei dat ze
 have.3SG.PRES INDEF aversion to hot weather and say.3SG.PST COMP she
 geen geld heeft.
 INDEF.NEG money have.3SG.PRES
 'Honestly, I don't know. I heard she went snorkeling in the Bahamas but I don't
 believe it. She hates hot weather and said she has no money.'

d. Turkish

Bahama-lar'da git-**miş**, ama bu ban-a inandırıcı gel-me-di
 Bahamas-PL.LOC go-**IND** but that I-DAT feasible come-NEG-PST
 'She went to the Bahamas (I heard), but I don't find it convincing.'

e. Japanese

Shoojikini-iu-to shira-nai. Bahama-ni shunookeringu-ni it-ta **sooda**
 honestly-say-COND know-NEG Bahamas-to snorkeling-to go-PST **REP**
 ga sore-ga jijitsu-da-to-wa omowa-nai. Kanojo-wa atsui-kikoo-ga
 but it-NOM fact-COP-COMP-TOP think-NEG she-TOP hot-weather-NOM
 kiraida si okane-ga nai-tte it-tei-ta.
 hate and money-NOM not.exist-COMP say-PROG-PST
 'Honestly speaking, I don't know. Apparently she went snorkeling in the Bahamas,
 but I don't think that it is the case. She hates hot weather and was saying she had no
 money.'

(43) is an instance of Speaker Disagreement, very similar to (41). The speaker signals that there is information that bears on the QUD (*Where did Ana go?*). Even though they personally disagree with it, they find it important to bring it up, along with specifying that they learned it via hearsay. (44) is a modification on the same theme and instantiates a case of Contrasting Viewpoints, similar to (42). As discussed in connection to example (30), two consecutive hearsay evidentials are often interpreted as coming from one source. Furthermore, as discussed in detail in Section 3, some evidentials do encode information about the speaker's belief about the prejacent, which will make them incompatible with (44).³² However, some evidentials can be used this way if it's marked explicitly that there were two distinct reports with contradicting information. In German, *sollen* can be modified with *erst* 'first' and *dann* 'then'. In Turkish, the second *miş* is modified with *de* 'also'.³³ In Japanese, sources are explicitly added via *niyoruto*-phrases, yielding a concord-like reading of the evidential. Unlike English *according to* adverbials, these expressions in Japanese don't seem to be intensional operators and the same sentences are odd without *sooda* (German often uses *laut X* phrases with the same function; they are extensional and merely signal the source of a report without shifting the world of evaluation).³⁴

32. Since such evidentials conventionally encode the speaker's belief, one cannot follow them with another evidential conveying an opposite belief.

33. Some people find such sentences degraded or at the very least unnatural. Even for them, *miş* is felicitous in (43).

34. Uses of hearsay evidentials that mark the source overtly are often called 'quotative' (Bary and Maier 2021; Krawczyk 2012), though there is no evidence that their semantics substantially differs from their other, source-

(44) **Contrasting Viewpoints**

After the winter break, we're discussing who went where on vacation. We're now talking about Ana. You haven't talked to her personally, but have heard various gossip.

- a. I heard she went snorkeling in the Bahamas. I also heard that she didn't. Oh well, you can't believe everything that you hear. (So I don't know.)
- b. #Apparently, she went snorkeling in the Bahamas. Apparently, she didn't go snorkeling.
- c. German
 Erst **soll** sie auf den Bahamas geschnorchelt haben. Dann
 first **REP.3SG.PRES** she at DEF Bahamas snorkel.PTCP have.INF then
soll sie nicht geschnorchelt haben.
REP.3SG.PRES she NEG snorkel.PTCP have.INF
 'First I heard she went snorkelling in the Bahamas, then I heard that she didn't.'
- d. Turkish
 Bahama-lar'da şnorkel yapmay-a git-**miş**. Şnorkel yapmay-a git-me-**miş** de.
 B-PL.LOC snorkel doing-DAT go-**IND** snorkel doing-DAT go-NEG-**IND** also
 'I heard she went snorkeling in the Bahamas. I also heard she didn't go snorkeling.'
- e. Japanese
 Shoojikini-iu-to shira-nai. John-niyoruto Bahama-ni shunookeringu-ni
 honestly-say-COND know-NEG John-according.to Bahamas-to snorkeling-to
 it-ta **sooda** ga Mary-niyoruto shunookeringu-ni-wa it-te-nai **sooda**.
 go-PST **REP** but Mary-according.to snorkeling-to-TOP go-PST-NEG **REP**
 kii-ta-koto subete-o shinji-ru-koto-wa deki-nai.
 hear-PST-thing all-ACC believe-PRES-NMLZ-TOP can-NEG.
 'Honestly speaking, I don't know. According to John, she went snorkeling in the Bahamas, but according to Mary, she didn't. You can't believe all you heard.'

The literature doesn't discuss the overall function of evidentials in discourses like (43) and (44), so it's not clear why the speaker chooses to present a piece of information they don't believe. I propose that (43) and (44) can be treated in parallel to (41) and (42), and that hearsay evidentials can behave like communicative predicates.

Hunter argues that non-endorsement with speech reports overrides the default according to which the main clause is interpreted as a parenthetical. As I have argued in Section 5.1, the evidential signal needn't be construed as always QUD-irrelevant. Instead, it is not-at-issue by default but can become at-issue when the discourse allows for it. Additionally, some evidentials require that their prejacent is always at-issue. In discourses like (43) and (44), we are presented with contrasting information. On the one hand, there is hearsay evidence that Ana went to the Bahamas. On the other hand, the speaker either disagrees with it based on their knowledge of the situation or has heard from another source that she didn't go. In both scenarios, the QUD is not only about the prejacent, but also about evidence (45).

(45) *Where did Ana go, and what evidence do we have for it?*

Apparently, she went snorkeling in the Bahamas, though I don't believe it, she hates the heat and she said she has no money.

unspecified uses or if such uses invoke the same mechanisms as actual quotation, e.g., if it makes the prejacent opaque for semantic operations.

I propose that speech acts in (43) and (44) should be assimilated to the treatment of declaratives with non-parenthetical uses of communicative predicates, however, we currently don't have a good framework for that.

Let's start with the existing theories of evidential meaning. They either don't model discourse dynamics at all or link evidentials to construction-specific discourse updates. Many modal approaches do not predict that hearsay evidence may be at-issue—for example, within Krazterian semantics modal bases are presupposed (see, e.g., recent discussion in [Roberts 2023](#)), and the dynamic account in ([McCready and Ogata 2007](#)) simply treats the hearsay operator as a test on an agent's information state (cf. [Veltman 1996](#) on *might*). Likewise, commitment-based approaches don't predict that evidential utterances can be associated with updates of different types. One could adopt [Koev's \(2022\)](#) proposal for parentheticals which allows both the host clause and the parenthetical to be at-issue, however, it runs afoul of non-endorsement, since the system treats all parenthetical constructions as committing the speaker to at least the possibility of the prejacent (cf. also discussion in Section 4.1). The account presented in Section 3.3 says nothing about at-issueness, so it is compatible with there being more than one at-issue contribution, but it runs into a more general problem.

Many current theories of conversational information flow, such as the Table Model, are not designed to handle complex discourse contributions of utterances with speech and attitude reports, since they usually take only one proposition to be at-issue at a given time. [Hunter \(2016\)](#) aims at redressing this gap, but her focus is the internal structure of larger discourses where different pieces of information are connected via coherence relations and not the more narrow question of assertion, so her framework cannot be easily applied to the updates of individual sentences. In order to fully capture the behavior of speech reports in a conversation we need a more flexible framework that (i) can track information attribution (such as, e.g., Mental State Discourse Representation Theory; [Kamp and Learnihan-Sylvester 2022](#)) and (ii) also makes an explicit reference to the QUD structure. This project takes us well beyond the scope of this paper, and I leave it for future research.

5.3 Weak endorsement as Relevance Implicature

Various forms of reported speech are known to trigger distancing effects, and it is still an open question whether such effects are conventionalized at least in some cases. The literature on evidentiality, on the other hand, often describes hearsay evidentials as giving rise to default endorsement. [AnderBois \(2014:244\)](#) claims that all hearsay evidentials are used almost veridically in the absence of disavowals. This is true for Cuzco Quechua *=si*, as [Faller \(2019\)](#) shows, but more work needs to be done to check how this works for other languages. AnderBois' overall generalization is clearly too strong, as it doesn't leave room for lexical variation. As we have already seen in Section 2.2, hearsay evidentials vary in whether they encode any information about the speaker's epistemic state, as diagnosed by the shape of available disavowals. Likewise, there is likely more variation with respect to endorsement and distancing (cf. [Wiemer and Socka 2017](#)). For example, Dutch *schijnen* is most naturally used when the source is unknown, which, in turn, may lead to inferences about doubt. To give another example, different constructions with a hearsay flavor differ in how frequently they appear in distancing contexts, including their co-occurrence with the word 'rumor' ([Mortelmans 2024, 2025](#)).

Setting potential variation in degrees of (non-)endorsement aside, my goal here is to make a weaker claim, which I will call WEAK ENDORSEMENT: in the absence of disavowals hearsay evidentials are interpreted in a way that the speaker is taken to have some credence in the preja-

cent.³⁵ I argue that weak endorsement arises as a Relevance Implicature with parenthetical uses of evidentials and is not present in their non-parenthetical uses, made possible by a different QUD (cf. [Benz and Jasinskaja 2017](#) for a discussion of the QUD-sensitivity of implicatures). The idea that endorsement is a type of implicature is not new. It has been mentioned in passing, and rejected, in ([Faller 2002](#)). It is also discussed by [Krawczyk \(2012:278\)](#) for evidentials and by [Hunter \(2016\)](#) for parenthetical speech reports. My innovation is to derive it specifically as a QUD-sensitive Relevance Implicature, building on the proposal that hearsay evidentials can have parenthetical and non-parenthetical uses.

Let us consider discourses with hearsay evidentials, again. I will illustrate everything with English *apparently* for the sake of readability, but the general reasoning is meant to apply to all hearsay evidentials that don't hard-wire endorsement. Let's start with a case of endorsement.

(46) *You learn via the grapevine that your friends are relocating.* [=35]

A: Why are you upset?

B: **Apparently**, my friends are moving away.

- a. B is cooperative and intends to address the QUD.
- b. The evidential signal is typically backgrounded and QUD-irrelevant.
- c. Some portion of an utterance must be QUD-relevant.
- d. The prejacent is QUD-relevant.³⁶
- e. B intends to resolve the QUD with the prejacent.
- f. B intends to update the common ground with the prejacent.
- g. B considers the prejacent a live possibility.

The reasoning in (46) is very much in line with what [Hunter \(2016\)](#) says about parenthetical uses of communicative predicates. In general, parenthetical constructions have been argued to be associated with some degree of endorsement. In [Murray's \(2014\)](#) system, this is directly encoded via construction-specific illocutionary relations. [Koev \(2022:108\)](#) treats it as an optimality calculation: since the speaker chose to use a construction whose prejacent has to be at-issue, they must intend to update the common ground with it and therefore must have some credence in it. My proposal for hearsay evidentials is slightly different, precisely because they have more flexibility than canonical parenthetical constructions.

In previous sections, I have argued that the evidential signal is QUD-irrelevant by default and that it may become relevant when the QUD structure allows for it. Utterances with disavowals are most natural in those discourses where not only the prejacent, but also the speaker's evidence for it becomes QUD-relevant. The overall discourse updates in those cases are more complex compared to ordinary assertions (hedged or unhedged) but crucially, the prejacent is not intended as the only update to the common ground. As such, the Relevance Implicature that is present in (46) fails to arise. Sometimes its absence may be signalled by an overt disavowal, as in the cases of Speaker Disagreement (43), but the same interpretation is also possible when the speaker is agnostic, as in the cases of Contrasting Viewpoints (44). Note that my approach can be seen as capturing the same intuition as [Faller's \(2019\)](#) appeal to the Cooperative Prin-

35. Note that this only applies to those evidentials that do not semantically encode that the speaker fully believes the prejacent; see Section 2.2.

36. Recall from Section 5.1 that not all evidentials mark their prejacent as obligatory at-issue. For those evidentials, the QUD-relevance of the prejacent is derived in conjunction with the QUD-irrelevance of the evidential signal.

ciple, but in more general terms. The key difference is that for Faller, the evidential signal is conventionally not-at-issue, while I have argued that it is conversationally not-at-issue in standard discourses and that it can become at-issue in specific contexts that override conversational defaults.³⁷

5.4 Recap

The literature takes the evidential signal to be conventionally not-at-issue. I propose instead that evidentials are best understood as conversationally not-at-issue, as determined by the QUD structure. Based on the refined answerhood diagnostic that pays attention to PREJACENT SETTLEDNESS, I have shown that some hearsay evidentials allow the evidential signal to be the main point of an utterance while some other ones require the prejacent to be at-issue (like, for example, slifting parentheticals). Importantly, this flexible approach predicts that hearsay evidentials will behave like communicative predicates and allow both parenthetical uses (only prejacent at-issue) and non-parenthetical uses (both prejacent and evidential signal at-issue). This prediction is borne out. Following [Hunter \(2016\)](#) on communicative predicates, I argue that discourses with overt disavowals have a special QUD structure that, unlike the more standard discourses, licenses non-parenthetical uses of evidentials. I further analyze WEAK ENDORSEMENT in the absence of disavowals as a Relevance Implicature. The main takeaway message is that hearsay evidentials might be best understood in comparison to discourse parentheticals and not slifting parentheticals, contra what was previously assumed.

6 Outlook

This paper offers a guided tour into the patterns of non-endorsement with hearsay evidentials by placing them in a broader context of speech reports (something that is rarely done, though see [Bary and Maier 2021](#); [Korotkova 2017](#)):

- it demonstrates that the behavior of hearsay vs. non-hearsay evidentials with respect to the possibility of endorsement reflects a more general distinction between speech vs. attitude reports: only speech reports allow it, so this property must be derived in a principled way;
- it shows that disavowals, often used to diagnose non-endorsement, aren't created equal and that their linguistic shape helps uncover previously unnoticed variation within hearsay evidentials with respect to the type of attitude they encode;
- it refines empirical diagnostics of evidential intensionality and shows that some evidentials behave like communicative predicates in that they license contradictions in their scope and block presuppositions, which is not predicted by extensional accounts;
- it refines empirical diagnostics of evidential not-at-issueness and argues that the evidential signal is conversationally not-at-issue, contrary to the received wisdom that is conventionally marked as not-at-issue;
- it draws attention to the discourse function of utterances with disavowals, largely overlooked in the existing literature, and proposes that such utterances address complex QUDs about both the prejacent and the available evidence, just like corresponding speech reports with communicative predicates.

37. As discussed by [Korotkova \(2020\)](#), Faller's proposal doesn't depend on treating the evidential signal as conventionally not-at-issue and her notion of at-issueness, though not the general framework of the Table model, is compatible with a more flexible approach along the lines advocated here.

The existing literature often assumes that hearsay evidentials are fundamentally different from communicative predicates in terms of their semantics and pragmatics. I propose a different take. On the one hand, it is true that evidentials differ from clause-taking verbs in a number of ways, including morphosyntactically—for example, they always express self-attributions, scope above negation and generally cannot express as many grammatical distinctions as full clauses.³⁸ On the other hand, this angle forces us to ignore the properties that evidentials do share with communicative predicates, such as the variation in the attitude they express (and possibly other dimensions of lexical variation) or their overall discourse dynamics. The debates about evidentiality tap into larger questions about modelling the relationship between discourse and grammar that surround, for example, clause typing, question bias or epistemic modality. Theories that model the illocutionary effects of evidentials often postulate construction-specific discourse updates, but at the same time it not entirely clear how corresponding discourses with communicative predicates should be modeled. In this paper, I specifically explore the discourses with disavowals. My larger point, echoing [Hunter \(2016\)](#), is that hearsay evidentials can have parenthetical and non-parenthetical uses and that we need a system that will generate these uses from their core semantics without hard-wiring the specific discourse updates they lead to.

Before I conclude, I would like to speculate on the relation between intensionality and at-issueness. Further support for the idea that hearsay evidentials have parenthetical and non-parenthetical uses comes from the behavior of communicative predicates in parenthetical constructions, which to the best of my knowledge has not been discussed in the previous literature. Like all clause-taking predicates, speech verbs are analyzed as involving intensional quantification. For example, they allow ‘de re’–‘de dicto’ ambiguities thus licensing contradictions in their scope (15a) and can block presuppositions (17). However, this only applies to their non-parenthetical uses. Intensional effects disappear when a speech predicate is used within an *as*-parenthetical (47) or a slifted clause (48). I illustrate this for English below, leaving detailed cross-linguistic work for future research.

(47) *As*-parentheticals

- a. #The deadliest virus, as the newspaper claimed, is not a virus.
- b. #Switzerland, as *die Zeit* reported, stopped drilling in the Arctic, though I don’t think they drilled there in the first place.

(48) Slifting parentheticals

- a. #The deadliest virus is not a virus, the newspaper claimed.
- b. #Switzerland stopped drilling in the Arctic, *die Zeit* reported, though I don’t think they drilled there in the first place.

This behavior is expected in multidimensional theories of parenthetical meaning in which there is no semantic or syntactic interaction between the host clause and the parenthetical ([Potts 2005](#) and later work). However, several recent theories argue that there is some degree of semantic interaction, for example, to allow anaphora ([Anderbois et al. 2015](#)) or commitment modulation ([Koev 2022](#)), and it is not entirely clear why the examples in (47) and (48) are

38. [Bary and Maier \(2021\)](#) argue that evidentials are non-eventive. This seems incorrect, given mounting evidence that many evidentials, including hearsay ones, make a reference to events of evidence acquisition in their semantics (see overview in [Hirayama and Matthewson 2022](#)). It might instead be more useful to think of evidentials as ‘light attitudes’ that have fewer categorial distinctions compared to full verbs ([Pancheva and Zubizarreta 2019](#); [Simeonova 2020](#)).

infelicitous. One could argue that the diagnostics I use for intensionality also require the lack of endorsement, but if endorsement isn't encoded directly, as Koev argues, it's unclear why (47) and (48) are ruled out. I suspect that something similar is at play with evidentials: they behave like regular intensional operators when the speaker doesn't endorse the prejacent (which, due to conversational defaults, we only see with overt disavowals that track the change in the QUD structure), but most of the time they behave like parentheticals, commenting on the speaker's source of information and possibly influencing the degree of their commitment to the prejacent. Theories that analyze speech acts with evidentials only model the latter use, while theories that treat evidentials as intensional operators typically don't discuss their update potential at all. I hope to see the reconciliation of those two views on evidentiality in future research.

References

- Aikhenvald, A. Y. (2004). *Evidentiality*. Oxford: Oxford University Press.
- Aikhenvald, A. Y. (2018). Evidentiality: The framework. In A. Y. Aikhenvald (Ed.), *The Oxford Handbook of Evidentiality*. Oxford: Oxford University Press.
- Anand, P., J. Grimshaw, and V. Hacquard (2017). Sentence embedding predicates, factivity and subjects. In C. Condoravdi (Ed.), *Lauri Karttunen Festschrift*. CSLI.
- Anand, P. and V. Hacquard (2013). Epistemics and attitudes. *Semantics and Pragmatics* 6(8), 1–59.
- Anand, P. and V. Hacquard (2014). Factivity, belief and discourse. In L. Crnić and U. Sauerland (Eds.), *The Art and Craft of Semantics: A Festschrift for Irene Heim*, pp. 69–90. Cambridge, MA: MITWPL.
- Anand, P. and N. Korotkova (2022). How to theorize about subjective meaning: A lesson from 'de re'. *Linguistics and Philosophy* 45(3), 619–681.
- AnderBois, S. (2014). On the exceptional status of reportative evidentials. In T. Snider, S. D'Antonio, and M. Weigand (Eds.), *Semantics and Linguistic Theory (SALT) 24*, pp. 234–254. LSA and CLC Publications.
- AnderBois, S. (2017). An illocutionary account of reportative evidentials in imperatives. In D. Burgdorf, J. Collard, S. Maspong, and B. Stefánsdóttir (Eds.), *Semantics and Linguistic Theory 27*, pp. 459–479.
- Anderbois, S., A. Brasoveanu, and R. Henderson (2015). At-issue proposals and appositive impositions in discourse. *Journal of Semantics* 32(1), 93–138.
- Bary, C. and N. Korotkova (2023). Expressing evidence. *Journal of Pragmatics* 218, 1–5.
- Bary, C. and E. Maier (2021). The landscape of speech reporting. *Semantics and Pragmatics* 14(8), 1–48.
- Benton, M. and P. van Elswyk (2020). Hedged assertion. In S. Goldberg (Ed.), *The Oxford Handbook of Assertion*, pp. 244–263. Oxford: Oxford University Press.
- Benz, A. and K. Jasinskaja (2017). Questions Under Discussion: From sentence to discourse. *Discourse Processes* 54(3), 177–186.
- Bergqvist, H. and K. Grzech (2023). The role of pragmatics in the definition of evidentiality. *STUF - Language Typology and Universals* 76(1), 1–30.
- Bhadra, D. (2020). The semantics of evidentials in questions. *Journal of Semantics* 37(3), 367–423.
- Bittner, M. (2011). Time and modality without tenses or modals. In R. Musan and M. Rathers (Eds.), *Tense across Languages*, pp. 147–188. Niemeyer.
- Boye, K. (2010). Semantic maps and the identification of cross-linguistic generic categories: Evidentiality and its relation to epistemic modality. *Linguistic Discovery* 8(1), 4–22.
- Brandom, R. (1983). Asserting. *Noûs* 17(4), 637–650.
- Chafe, W. and J. Nichols (Eds.) (1986). *Evidentiality: The Linguistic Coding of Epistemology*. Norwood: Ablex Publishing Corporation.

- Davis, C., C. Potts, and M. Speas (2007). The pragmatic values of evidential sentences. In T. Friedman and M. Gibson (Eds.), *Semantics and Linguistic Theory 17*, Ithaca, NY, pp. 71–88. Cornell University.
- DeRose, K. (1991). Epistemic possibilities. *Philosophical Review* 100, 581–605.
- Diewald, G. and E. Smirnova (2010). Introduction. Evidentiality in European languages: The lexical-grammatical distinction. In G. Diewald and E. Smirnova (Eds.), *Linguistic Realization of Evidentiality in European Languages*, pp. 1–14. Berlin/New York, NY: De Gruyter Mouton.
- Djäv, K. (2023). Knowing and believing things: What DP-complements can tell us about the argument structure and composition of (factive) attitudes. *Journal of Semantics* 40(2), 1–55.
- Doron, E. (1991). Point of view as a factor of content. In S. Moore and A. Wyner (Eds.), *Proceedings of Semantics and Linguistic Theory 1*, pp. 51–64. Cornell University.
- Dorst, K. and M. Mandelkern (2023). Good guesses. *Philosophy and Phenomenological Research* 105(3), 581–618.
- Dudley, R. and C. Vogel (2023). Knowledge, the concept KNOW, and the word know: considerations from polysemy and pragmatics. *Synthese* 203(1).
- Eckardt, R. (2014). *The Semantics of Free Indirect Discourse: How Texts Allow Us to Mind-read and Eavesdrop*. Leiden: Brill.
- Fabricius-Hansen, C. and K. J. Sæbø (2004). In a mediative mood: The semantics of the german reportive subjunctive. *Natural Language Semantics* 12(3), 213–257.
- Faller, M. (2002). *Semantics and pragmatics of evidentials in Cuzco Quechua*. Ph. D. thesis, Stanford University.
- Faller, M. (2007). Evidentiality above and below speech acts. Ms., University of Manchester.
- Faller, M. (2019). The discourse commitments of illocutionary reportatives. *Semantics and Pragmatics* 12(8), 1–46.
- Faller, M. (2023). Conjectural speech acts in Cuzco Quechua. *Journal of Pragmatics* 214, 144–163.
- Faller, M. (2024). The interrogative flip with illocutionary evidentials. *Folia Linguistica* 59(1), 175–205.
- Farkas, D. (1997). Evaluation indices and scope. In A. Szabolcsi (Ed.), *Ways of Scope Taking*, pp. 183–215. Dordrecht: Springer Netherlands.
- Farkas, D. and K. Bruce (2010). On reacting to assertions and polar questions. *Journal of Semantics* 27(1), 81–118.
- von Fintel, K. and A. S. Gillies (2010). Must ... stay ... strong! *Natural Language Semantics* 18(4), 351–383.
- von Fintel, K. and A. S. Gillies (2011). ‘Might’ made right. In A. Egan and B. Weatherston (Eds.), *Epistemic Modality*, pp. 108–130. Oxford University Press.
- von Fintel, K. and A. S. Gillies (2021). Still going strong. *Natural Language Semantics* 29, 91–113.
- von Fintel, K. and I. Heim (2011). Intensional semantics. MIT Lecture notes, <http://web.mit.edu/fintel/fintel-heim-intensional.pdf>.
- Forker, D. (2018). Evidentiality and its relations with other verbal categories. In A. Aikhenvald (Ed.), *The Oxford Handbook of Evidentiality*, pp. 65–84. Oxford: Oxford University Press.
- Gajewski, J. (2007). Neg-raising and polarity. *Linguistics and Philosophy* 30, 289–328.
- Gillies, A. S. (2001). A new solution to Moore’s paradox. *Philosophical Studies* 105(3), 237–250.
- Goffman, E. (1979). Footing. *Semiotica* 25(1-2), 1–30.
- Goldberg, S. (Ed.) (2020). *The Oxford Handbook of Assertion*. Oxford: Oxford University Press.
- Grimshaw, J. (2015). The light verbs Say and SAY. In I. Toivonen, P. Csúri, and E. van der Zee (Eds.), *Structures in the Mind: Essays on Language, Music, and Cognition*, pp. 79–99. Cambridge: MIT Press.
- Grzech, K. and H. Bergqvist (Eds.) (2025). *Expanding the Boundaries of Epistemicity*. Berlin: De Gruyter.

- Gunlogson, C. (2003). *True to Form: Rising and Falling Declaratives as Questions in English*. New York: Routledge.
- de Haan, F. (2013a). Coding of evidentiality. In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology.
- de Haan, F. (2013b). Semantic distinctions of evidentiality. In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology.
- Hintikka, J. (1969). Semantics for propositional attitudes. In J. Davis, D. Hockney, and W. Wilson (Eds.), *Philosophical Logic*, pp. 21–45. Dordrecht: Reidel.
- Hirayama, Y. and L. Matthewson (2022). Evidential-temporal interactions do not (always) come for free. *Journal of Pragmatics* 193, 173–188.
- Hunter, J. (2016). Reports in discourse. *Dialogue and Discourse* 7(4), 1–35.
- Hunter, J. and M. Abrusán (2017). Rhetorical structure and QUDs. In M. Otake, S. Kurahashi, Y. Ota, K. Satoh, and D. Bekki (Eds.), *New Frontiers in Artificial Intelligence*, pp. 41–57. Cham: Springer.
- Izvorski, R. (1997). The present perfect as an epistemic modal. In A. Lawson (Ed.), *Semantics and Linguistic Theory (SALT) 7*, Ithaca, NY, pp. 222–239. LSA and CLC Publications.
- Johnson, K. (2022). Time and evidence in the graded tense system of Mvskoke (Creek). *Natural Language Semantics* 30(2), 155–183.
- Kamp, H. and B. Learnihan-Sylvester (2022). Introduction to MSDRT. Ms.
- Kaufmann, M. (2014). Embedded imperatives across languages: Too rare to expect, too frequent to ban. Handout for presentation at Colloquium Stony Brook, April 4, 2014.
- Kaufmann, M. and S. Kaufmann (2020). Talking about sources. In M. Asatryan, Y. Song, and A. Whitmal (Eds.), *NELS 50: Proceedings of the Fiftieth Annual Meeting of the North East Linguistic Society*, Amherst, Massachusetts. GLSA.
- Kaufmann, M. and C. Poschmann (2013). Embedded imperatives: Empirical evidence from Colloquial German. *Language* 89(3), 619–637.
- Keshet, E. (2010). Split intensionality: A new scope theory of *de re* and *de dicto*. *Linguistics and Philosophy* 33(4), 251–283.
- Kierstead, G. (2015). Projectivity and the Tagalog Reportative Evidential. MA Thesis, Ohio State University, Columbus.
- Koev, T. (2018). Notions of at-issueness. *Language and Linguistics Compass* 12(12), 1–16.
- Koev, T. (2019). Strong beliefs, weak commitments. In T. M. Espinal (Ed.), *Proceedings of Sinn und Bedeutung 23*, Universitat Autònoma de Barcelona, Bellaterra (Cerdanyola del Vallès), pp. 557–574.
- Koev, T. (2022). *Parenthetical Meaning*. Oxford: Oxford University Press.
- Koev, T. (2025). An argument for a strong force semantics of *believe*. *Linguistic Inquiry*. Advanced publication.
- Koring, L. (2013). *Seemingly similar: Subjects and displacement in grammar, processing, and acquisition*. Ph. D. thesis, Utrecht University.
- Korotkova, N. (2012). Evidentiality in the Georgian tense and aspect system. Ms., UCLA.
- Korotkova, N. (2015). Evidentials in attitudes: do’s and don’t’s. In E. Csipak and H. Zeijlstra (Eds.), *Sinn und Bedeutung 19*, pp. 340–357.
- Korotkova, N. (2016a). Disagreement with evidentials: A call for subjectivity. In J. Hunter, M. Simons, and M. Stone (Eds.), *JerSem: The 20th Workshop on the Semantics and Pragmatics of Dialogue*, pp. 65–75.
- Korotkova, N. (2016b). *Heterogeneity and Universality in the Evidential Domain*. Ph. D. thesis, University of California, Los Angeles.

- Korotkova, N. (2017). Evidentials and (relayed) speech acts: Hearsay as quotation. In S. D’Antonio, M. Moroney, and C. R. Little (Eds.), *Semantics and Linguistic Theory (SALT) 25*, pp. 676–694. LSA Open Journal Systems.
- Korotkova, N. (2020). Evidential meaning and (not-)at-issueness. *Semantics & Pragmatics* 13(4), 1–24.
- Korotkova, N. (2021). The embedding puzzle: Constraints on evidentials in complement clauses. *Linguistic Inquiry* 52(1), 210–226.
- Korotkova, N. (2026). The notional category of evidentiality. Submitted to: Arregui, A., V. Hacquard and M. Ippolito (Eds.), *The Cambridge Handbook of Modality*.
- Korotkova, N. and P. Anand (2025). Nothing to claim: *Claim* reports and non-endorsement. In E. Csipak, J. David, and M. Liu (Eds.), *Festschrift in Honour of Regine Eckardt*, pp. 139–148. Berlin: Humboldt-Universität zu Berlin.
- Kratzer, A. (1977). What ‘Must’ and ‘Can’ must and can mean. *Linguistics and Philosophy* 1, 337–355.
- Kratzer, A. (1981). The notional category of modality. In H.-J. Eikmeyer and H. Rieser (Eds.), *Words, Worlds, and Contexts*, pp. 38–74. de Gruyter.
- Kratzer, A. (1991). Modality. In A. von Stechow and D. Wunderlich (Eds.), *Semantics: An International Handbook of Contemporary Research*, pp. 639–650. de Gruyter.
- Kratzer, A. (2002). Facts: Particulars or information units? *Linguistics and Philosophy* 25, 655–670.
- Kratzer, A. (2012). *Modals and Conditionals*. Oxford: Oxford University Press.
- Krawczyk, E. A. (2012). *Inferred Propositions and the Expression of the Evidence Relation in Natural Language. Evidentiality in Central Alaskan Yup’ik Eskimo and English*. Ph. D. thesis, Georgetown University.
- Krifka, M. (2019). Commitments and beyond. *Theoretical Linguistics* 45(1-2), 73–91.
- Lasnik, P. (2005). Context dependence, disagreement, and predicates of personal taste. *Linguistics and Philosophy* 28(6), 643–686.
- Lassiter, D. (2016). *Must*, knowledge and (in)directness. *Natural Language Semantics* 24(2), 117–163.
- Lawlor, K. and J. Perry (2008). Moore’s paradox. *Australasian Journal of Philosophy* 86(3), 421–427.
- Lee, J. (2013). Temporal constraints on the meaning of evidentiality. *Natural Language Semantics* 21(1), 1–41.
- MacFarlane, J. (2011). Epistemic modals are assessment-sensitive. In A. Egan and B. Weatherson (Eds.), *Epistemic modality*, Chapter 5, pp. 144–178. Oxford University Press.
- Maier, E. (2014). Mixed quotation: The grammar of apparently transparent opacity. *Semantics and Pragmatics* 7, 1–67.
- Malamud, S. and T. Stephenson (2015). Three ways to avoid commitments: Declarative force modifiers in the conversational scoreboard. *Journal of Semantics* 32(2), 275–311.
- Marsili, N. (2024). The definition of assertion: Commitment and truth. *Mind & Language* 39(4), 540–560.
- Martínez Vera, G. (2020). *On the Semantics of Evidentials in Southern Aymara*. Ph. D. thesis, University of Connecticut.
- Martínez Vera, G. (2024). Direct evidentiality and discourse in Southern Aymara. *Natural Language Semantics* 32, 1–34.
- Matsubara, J. (2017). *The Semantics and Pragmatics of the Japanese Evidentials -Rashii, -Sooda, and -Yooda: an Experimental Investigation*. Ph. D. thesis, Northwestern University.
- Matthewson, L., H. Davis, and H. Rullman (2007). Evidentials as epistemic modals: Evidence from St’át’imcets. In J. van Craenenbroeck (Ed.), *Linguistic Variation Yearbook* 7, pp. 201–254. Amsterdam: John Benjamins.
- McCready, E. (2015). *Reliability in Pragmatics*. Oxford: Oxford University Press.

- McCready, E. and N. Ogata (2007). Evidentiality, modality and probability. *Linguistics and Philosophy* 30(2), 147–206.
- Merigli, B. (2016). Modeling Indirect Evidence. Master's thesis, University of California, Santa Cruz.
- Mortelmans, T. (2000). On the 'evidential' nature of the 'epistemic' use of the German modals *müssen* and *sollen*. *Belgian Journal of Linguistics* 14, 131–148.
- Mortelmans, T. (2024). Frequency differences in reportative exceptionality and how to account for them: A case study on verbal reportative markers in French, Dutch and German. *Studies in Language* 48(3), 682–722.
- Mortelmans, T. (2025). How do 'rumours' and reportative evidentiality match? A comparative study of the French conditional, Dutch *zou* + INF, and German *sollen* + INF. *Journal of Linguistics*, 1–37.
- Murray, S. E. (2010). *Evidentiality and the Structure of Speech Acts*. Ph. D. thesis, Rutgers University, New Brunswick, NJ.
- Murray, S. E. (2014). Varieties of update. *Semantics and Pragmatics* 7(2), 1–53.
- Murray, S. E. (2017). *The Semantics of Evidentials*. Oxford: Oxford University Press.
- Murray, S. E. (2020). Evidentials. In L. Matthewson, C. Meier, H. Rullman, and T. E. Zimmerman (Eds.), *The Blackwell Companion to Semantics*. Wiley.
- Murray, S. E. (2021). Evidentiality, modality, and speech acts. *Annual Review of Linguistics* 7(1), 213–233.
- Musan, R. (1997). *On the Temporal Interpretation of Noun Phrases*. New York: Garland.
- Nagel, J. (2017). Factive and nonfactive mental state attribution. *Mind & Language* 32(5), 525–544.
- Northrup, O. (2014). *Grounds for commitment*. Ph. D. thesis, University of California, Santa Cruz.
- Pagin, P. and N. Marsili (2025). Assertion. In E. N. Zalta and U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Fall 2025 ed.). Metaphysics Research Lab, Stanford University.
- Pancheva, R. and D. Rudin (2019). Speaker discourse roles and the discourse profile of reportative evidentials. In J. Schlöder, D. McHugh, and F. Roelofsen (Eds.), *Proceedings of the 22nd Amsterdam Colloquium*, pp. 327–336.
- Pancheva, R. and M. L. Zubizarreta (2019). On the role of person features in the evidential-temporal connection. *The Canadian Journal of Linguistics* 64(4), 673–708.
- Papafragou, A. (2006). Epistemic modality and truth conditions. *Lingua* 116(10), 1688–1702.
- Peterson, T. (2010). *Epistemic Modality and Evidentiality in Gitksan at the Semantics-Pragmatics Interface*. Ph. D. thesis, University of British Columbia.
- Phillips, J., W. Buckwalter, F. Cushman, O. Friedman, A. Martin, J. Turri, L. Santos, and J. Knobe (2021). Knowledge before belief. *Behavioral and Brain Sciences* 44.
- Potts, C. (2005). *The Logic of Conventional Implicatures*. Oxford: Oxford University Press.
- Roberts, C. (2023). The indexical character of epistemic modality. *Linguistics and Philosophy* 46(5), 1219–1267.
- Roberts, T. (2019). I can't believe it's not lexical: Deriving distributed veridicality. In K. Blake, F. Davis, K. Lamp, and J. Rhyne (Eds.), *Semantics and Linguistic Theory* 29, pp. 665–685. Linguistic Society of America.
- Rooryck, J. (2001a). Evidentiality, part I. *Glott International* 5(4), 125–133.
- Rooryck, J. (2001b). Evidentiality, part II. *Glott International* 5(5), 161–168.
- Ross, J. R. (1973). Slifting. In M. Gross, M. Halle, and M.-P. Schützenburger (Eds.), *The Formal Analysis of Natural Languages: Proceedings of the First International Conference, Paris, April 27–29, 1970*, pp. 133–169. Berlin, Boston: De Gruyter.
- Rudin, D. (2022). Intonational commitments. *Journal of Semantics* 39(2), 339–383.
- Rudin, D. (2025a). Asserting epistemic modals. *Linguistics and Philosophy* 48(1), 43–88.

- Rudin, D. (2025b). Embedded intonation and quotative complements to verbs of speech. *Linguistic Inquiry*, 1–60.
- Scheffler, T. and S. Malamud (2023). “Won’t you?” Reverse-polarity question tags in American English as a window into the semantics-pragmatics interface. *Linguistics and Philosophy* 46, 1285–1327.
- Schenner, M. (2010a). Embedded evidentials in German. In G. Diewald and E. Smirnova (Eds.), *Linguistic realization of evidentiality in European languages*, pp. 157–185. Mouton de Gruyter.
- Schenner, M. (2010b). Evidentials in complex sentences: Foundational issues and data from German and Turkish. In T. Peterson and U. Sauerland (Eds.), *Evidence from Evidentials*, pp. 183–220. Vancouver: University of British Columbia.
- Schwager, M. (2010). On what has been said in Tagalog: Reportative *daw*. In T. Peterson and U. Sauerland (Eds.), *Evidence from Evidentials*, pp. 221–246. Vancouver: University of British Columbia.
- Schwarz, F. (2012). Situation pronouns in determiner phrase. *Natural Language Semantics* 20(4), 431–475.
- Semeijn, M. (2024). On the difference between the ‘In’ and ‘According to’ operators. *Linguistics and Philosophy* 47(2), 239–264.
- Şener, N. (2011). *Semantics and Pragmatics of Evidentials in Turkish*. PhD dissertation, University of Connecticut, Storrs.
- Sharvit, Y. (2008). The puzzle of free indirect discourse. *Linguistics and Philosophy* 31(3), 353–395.
- Simeonova, V. (2020). *The Syntax and Semantics of Light Attitudes*. Ph. D. thesis, University of Ottawa.
- Simons, M. (2007). Observations on embedding verbs, evidentiality, and presupposition. *Lingua* 117(6), 1034–1056.
- Simons, M., D. Beaver, C. Roberts, and J. Tonhauser (2017). The best question: Explaining the projection behavior of factives. *Discourse Processes* 54(3), 187–206.
- Simons, M., J. Tonhauser, D. Beaver, and C. Roberts (2010). What projects and why. In N. Li and D. Lutz (Eds.), *Semantics and Linguistic Theory (SALT) 20*, Ithaca, NY, pp. 309–327. LSA and CLC Publications.
- Slobin, D. and A. Akşu (1982). Tense, aspect, and modality in the use of the Turkish evidentials. In P. Hopper (Ed.), *Tense-Aspect: Between Semantics and Pragmatics*, pp. 185–200. Benjamins.
- Smirnova, A. (2012). Evidentiality in Bulgarian: Temporality, epistemic modality, and information source. *Journal of Semantics* 30, 479–532.
- Snider, T. (2017). *Anaphoric Reference to Propositions*. Ph. D. thesis, Cornell University.
- Stalnaker, R. (2000). On “Moore’s paradox”. In P. Engel (Ed.), *Believing and Accepting*, pp. 93–100. Dordrecht: Springer.
- Stec, K., M. Huiskes, and G. Redeker (2016). Multimodal quotation: Roleshift practices in spoken narratives. *Journal of Pragmatics* 104, 1–17.
- Thomas, G. (2014). Embedded imperatives in Mbyá. In H.-L. Huang, E. Poole, and A. Rysling (Eds.), *43rd Annual Meeting of the North East Linguistic Society (NELS)*, pp. 181–194. CreateSpace.
- Tonhauser, J., D. Beaver, C. Roberts, and M. Simons (2013). Towards a taxonomy of projective content. *Language* 89(1), 66–109.
- van Elswyk, P. (2025). Veridicality and the acquisition of *think*. *Linguistics and Philosophy* 48(2), 353–370.
- Veltman, F. (1996). Defaults in update semantics. *Journal of Philosophical Logic* 25(3), 221–261.
- Vetter, B. (2011). Recent work: Modality without possible worlds. *Analysis* 71(4), 742–754.
- Waldie, R. J. (2013). *Evidentiality in Nuu-chah-nulth*. Ph. D. thesis, University of British Columbia.
- Walker, M. A. (1996). Inferring acceptance and rejection in dialog by default rules of inference. *Language and Speech* 39(2-3), 265–304.

- Weatherson, B. and A. Egan (2011). Introduction: Epistemic modals and epistemic modality. In A. Egan and B. Weatherson (Eds.), *Epistemic Modality*, pp. 1–18. Oxford: Oxford University Press.
- Wiemer, B. and A. Socka (2017). How much does pragmatics help to contrast the meaning of hearsay adverbs? (part 1). *Studies in Polish Linguistics* 12(1), 27–56.
- Willer, M. (2013). Dynamics of epistemic modality. *The Philosophical Review* 122(1), 45–92.
- Willer, M. (2015). An update on epistemic modals. *Journal of Philosophical Logic* 44(6), 835–849.
- Willett, T. (1988). A cross-linguistic survey of the grammaticization of evidentiality. *Studies in Language* 12(1), 51–97.
- Yalcin, S. (2007). Epistemic modals. *Mind* 116(464), 983–1026.
- Yalcin, S. (2011). Nonfactualism about epistemic modality. In A. Egan and B. Weatherson (Eds.), *Epistemic Modality*, Chapter 10, pp. 295–332. Oxford University Press.